

PRACTICAL NO:-01

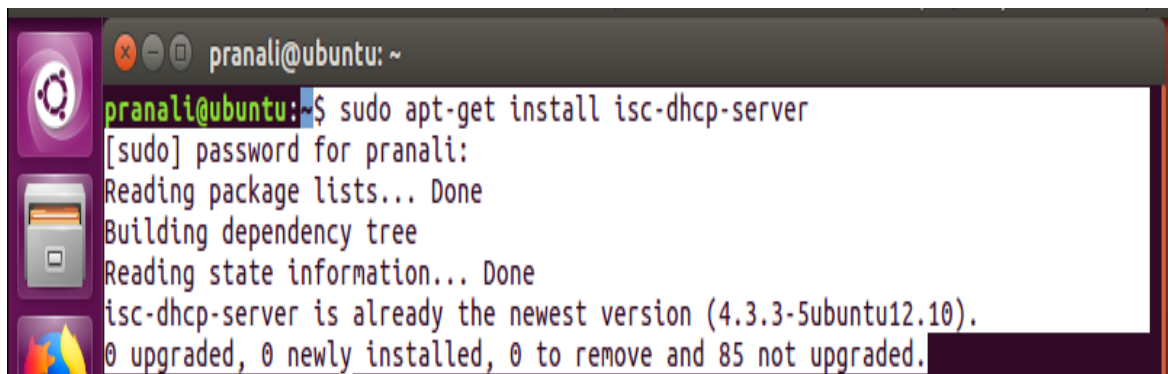
AIM:-

Install DHCP Server in Ubuntu 16.04. Configure DHCP Server, Configure DHCP (Dynamic Host Configuration Protocol) Server.

COMMAND:-

Sudo apt-get install isc-dhcp-server

OUTPUT:-

A screenshot of a Linux terminal window. The window title is 'pranali@ubuntu: ~'. The terminal shows the command 'pranali@ubuntu:~\$ sudo apt-get install isc-dhcp-server' being entered. The output of the command is: '[sudo] password for pranali:', 'Reading package lists... Done', 'Building dependency tree', 'Reading state information... Done', 'isc-dhcp-server is already the newest version (4.3.3-5ubuntu12.10).', and '0 upgraded, 0 newly installed, 0 to remove and 85 not upgraded.' The terminal has a dark background with light-colored text. On the left side of the terminal window, there are three icons: a gear, a server rack, and a flame.

PRACTICAL NO:-02

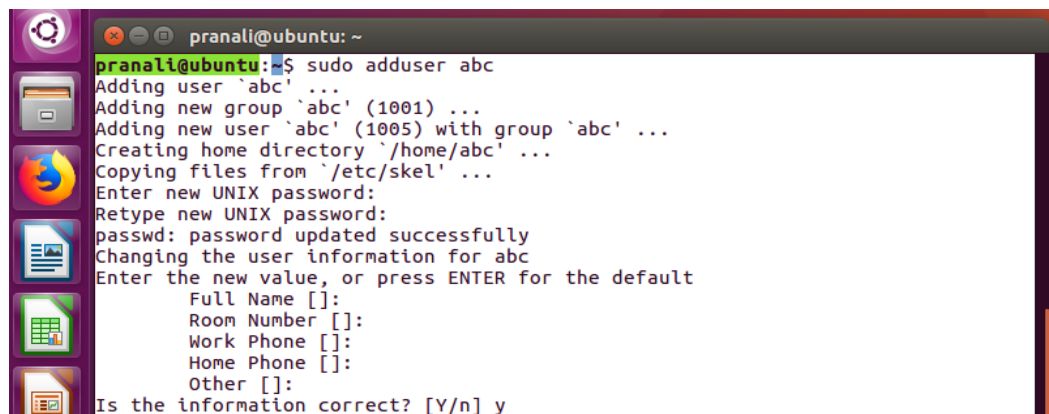
AIM:-

Initial settings: Add a User, Network Settings, Change to static IP address, Disable IPv6 if not needed, Configure Services, display the list of services which are running Sudo Settings

COMMAND:-

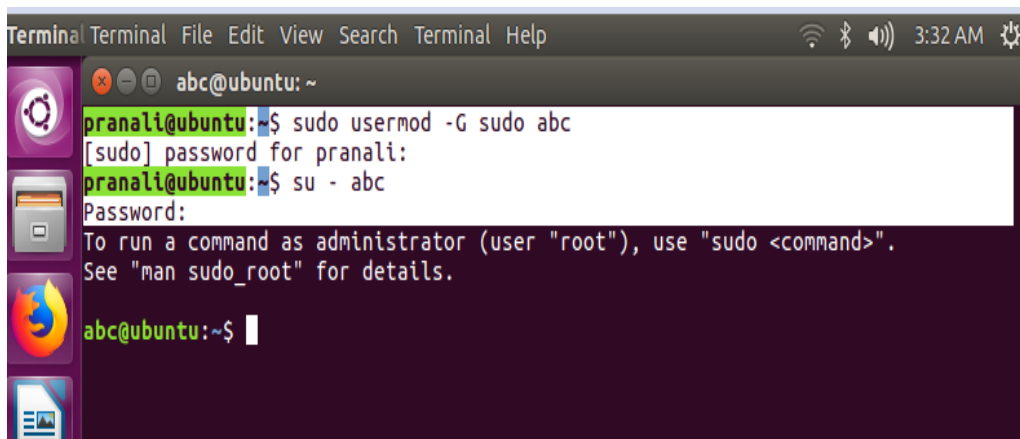
Add user

- 1) `sudo adduser abc`



```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo adduser abc  
Adding user `abc' ...  
Adding new group `abc' (1001) ...  
Adding new user `abc' (1005) with group `abc' ...  
Creating home directory `/home/abc' ...  
Copying files from `/etc/skel' ...  
Enter new UNIX password:  
Retype new UNIX password:  
passwd: password updated successfully  
Changing the user information for abc  
Enter the new value, or press ENTER for the default  
Full Name []:  
Room Number []:  
Work Phone []:  
Home Phone []:  
Other []:  
Is the information correct? [Y/n] y
```

- 2) `sudo useradd -G sudo abc`
- 3) `su - abc`

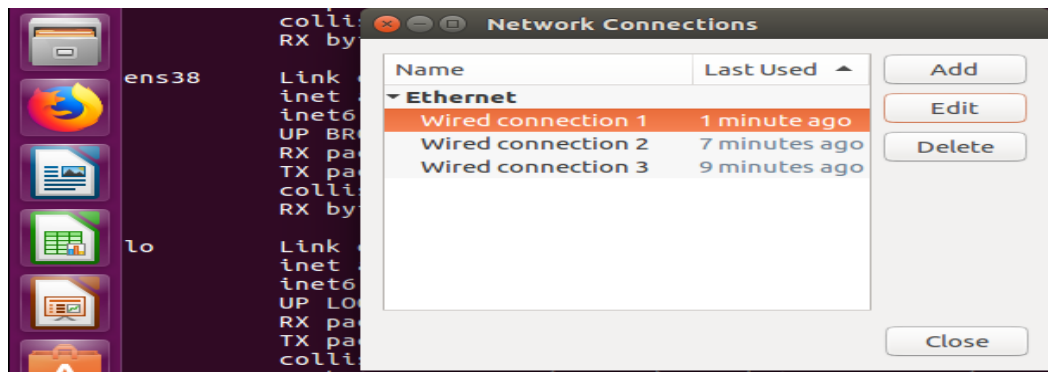


```
Terminal Terminal File Edit View Search Terminal Help  
abc@ubuntu: ~  
pranali@ubuntu:~$ sudo usermod -G sudo abc  
[sudo] password for pranali:  
pranali@ubuntu:~$ su - abc  
Password:  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
abc@ubuntu:~$
```

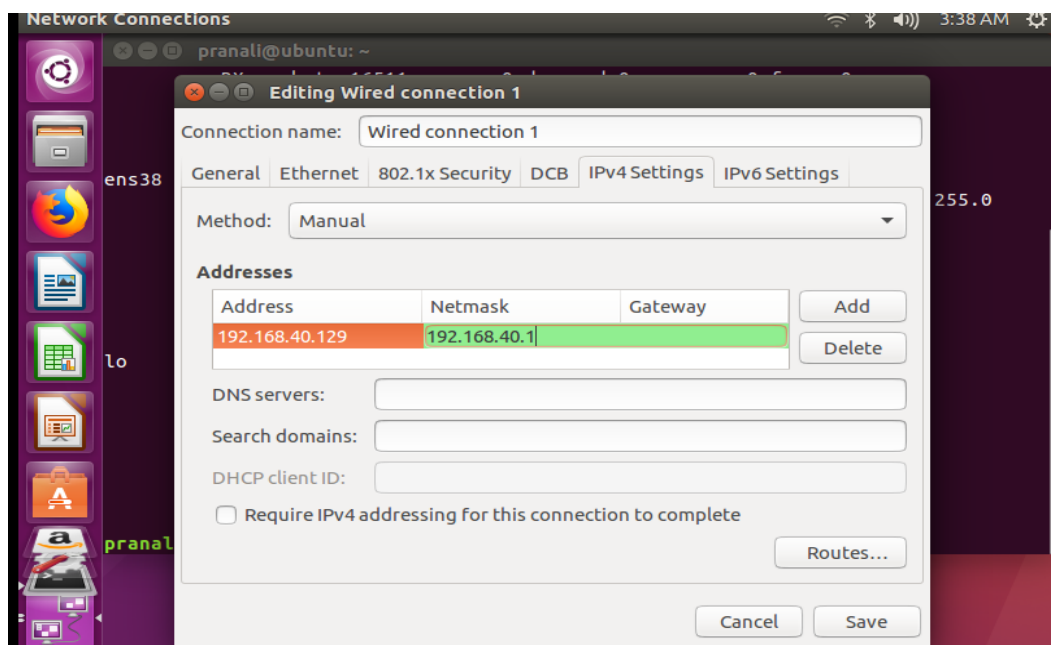
- 4) Reboot the system.

LINUX SERVER ADMINISTRATION

Network setting

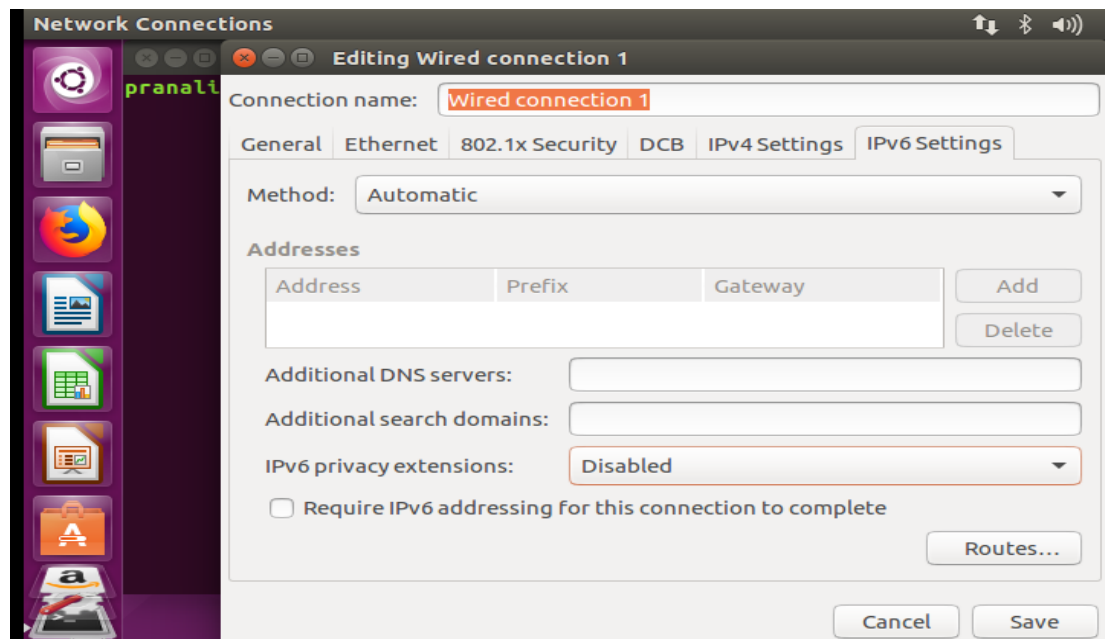


Gives the ip address default gateway

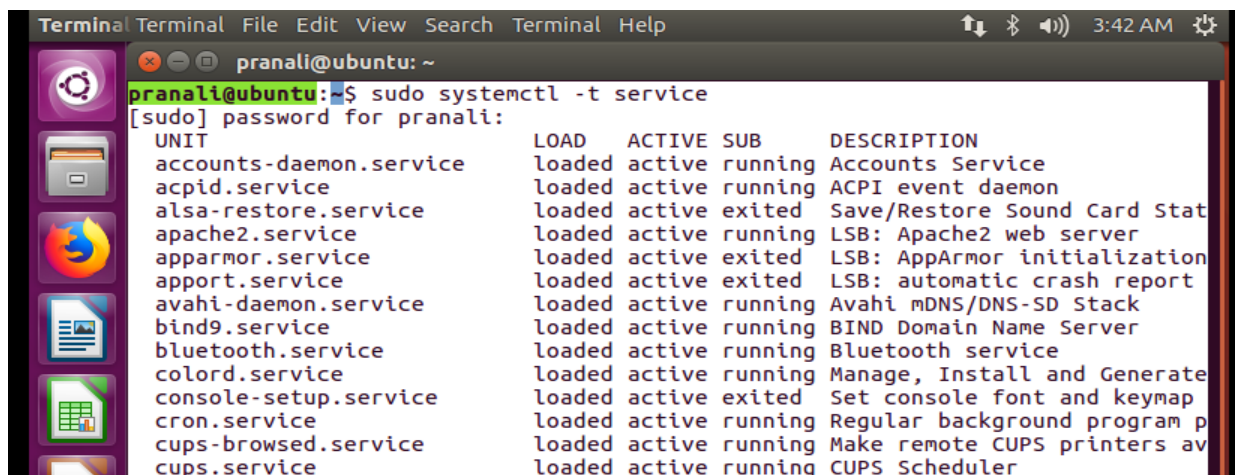


LINUX SERVER ADMINISTRATION

Disable ipv6



List out the all service and configuration service



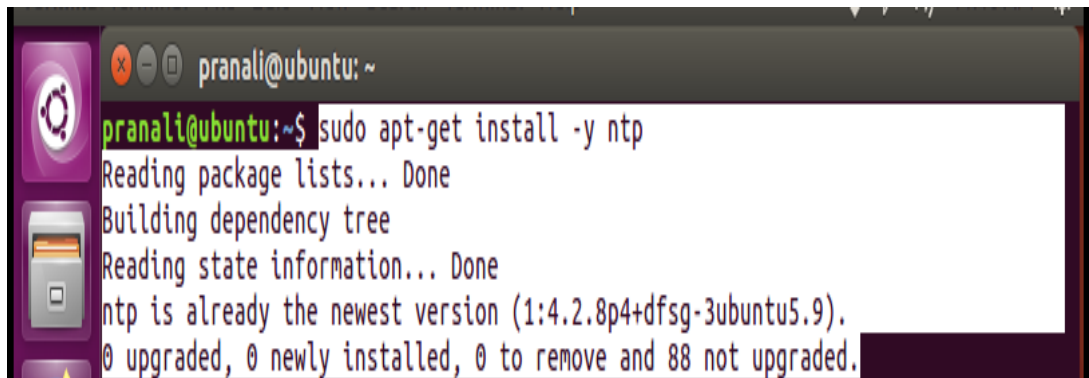
PRACTICAL NO:-03

AIM:-

Configure NTP Server (NTPd), Install and Configure NTPd, Configure NTP Client (Ubuntu and Windows)

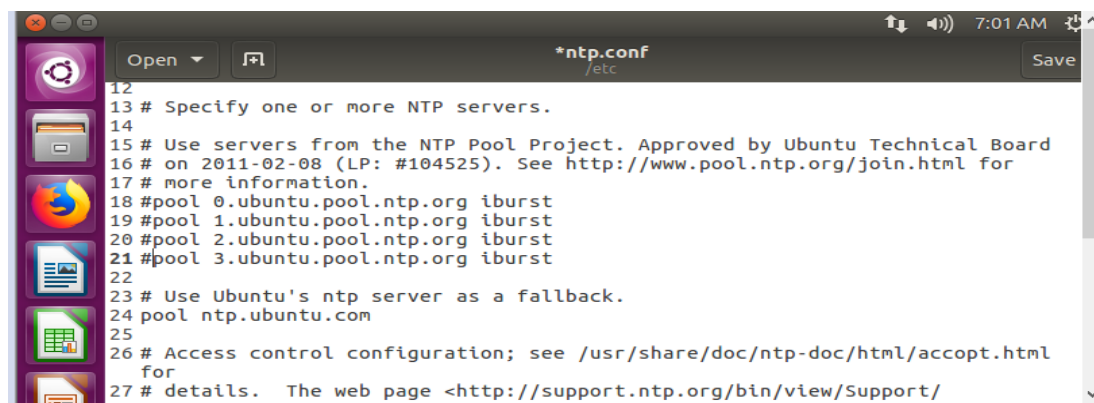
COMMAND:-

- 1) `sudo apt-get install -y ntp`



```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install -y ntp  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
ntp is already the newest version (1:4.2.8p4+dfsg-3ubuntu5.9).  
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
```

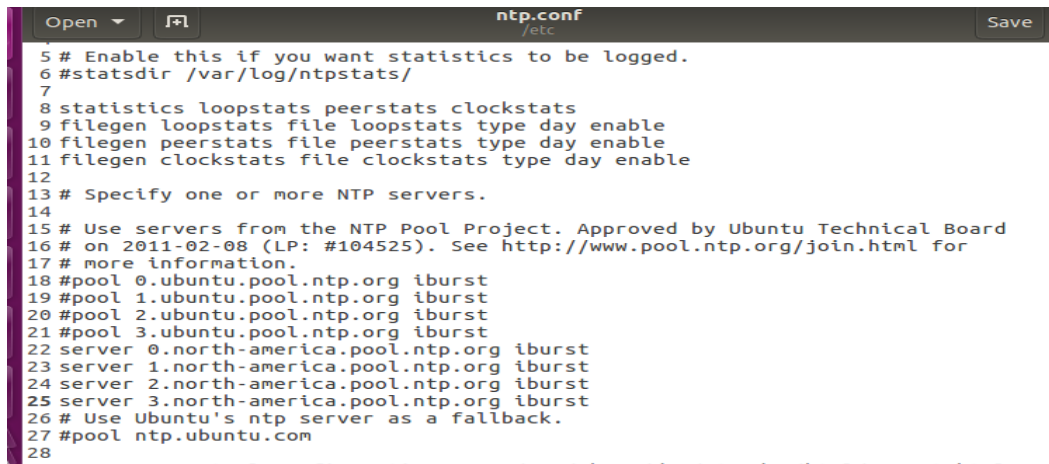
- 2) `sudo gedit /etc/ntp.conf`



```
*ntp.conf  
/etc  
12  
13 # Specify one or more NTP servers.  
14  
15 # Use servers from the NTP Pool Project. Approved by Ubuntu Technical Board  
16 # on 2011-02-08 (LP: #104525). See http://www.pool.ntp.org/join.html for  
17 # more information.  
18 #pool 0.ubuntu.pool.ntp.org iburst  
19 #pool 1.ubuntu.pool.ntp.org iburst  
20 #pool 2.ubuntu.pool.ntp.org iburst  
21 #pool 3.ubuntu.pool.ntp.org iburst  
22  
23 # Use Ubuntu's ntp server as a fallback.  
24 pool ntp.ubuntu.com  
25  
26 # Access control configuration; see /usr/share/doc/ntp-doc/html/accpt.html  
27 # details. The web page <http://support.ntp.org/bin/view/Support/
```

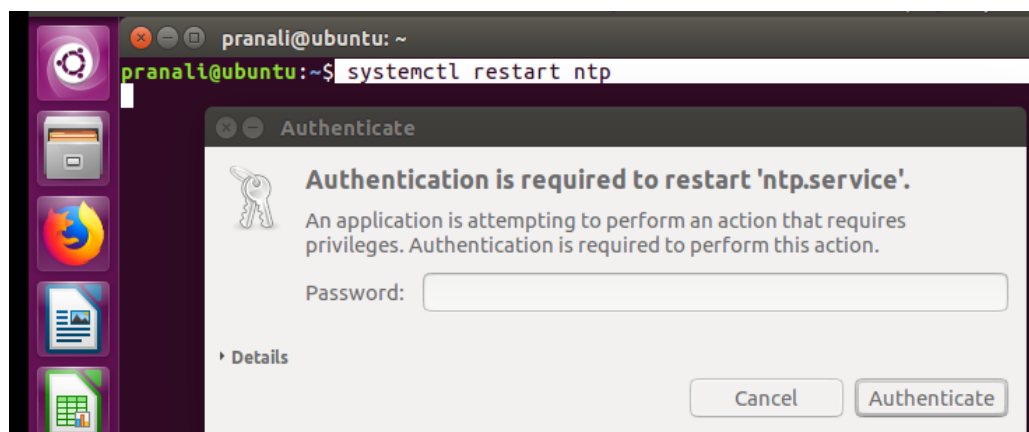
- 3) comment the line number 18 to 21.
Eg. `#pool 0.ubuntu.pool.ntp.org iburst`
And write the following lines:-
`server 0.north-america.pool.ntp.org iburst`
`server 1.north-america.pool.ntp.org iburst`
`server 2.north-america.pool.ntp.org iburst`
`server 3.north-america.pool.ntp.org iburst`

LINUX SERVER ADMINISTRATION

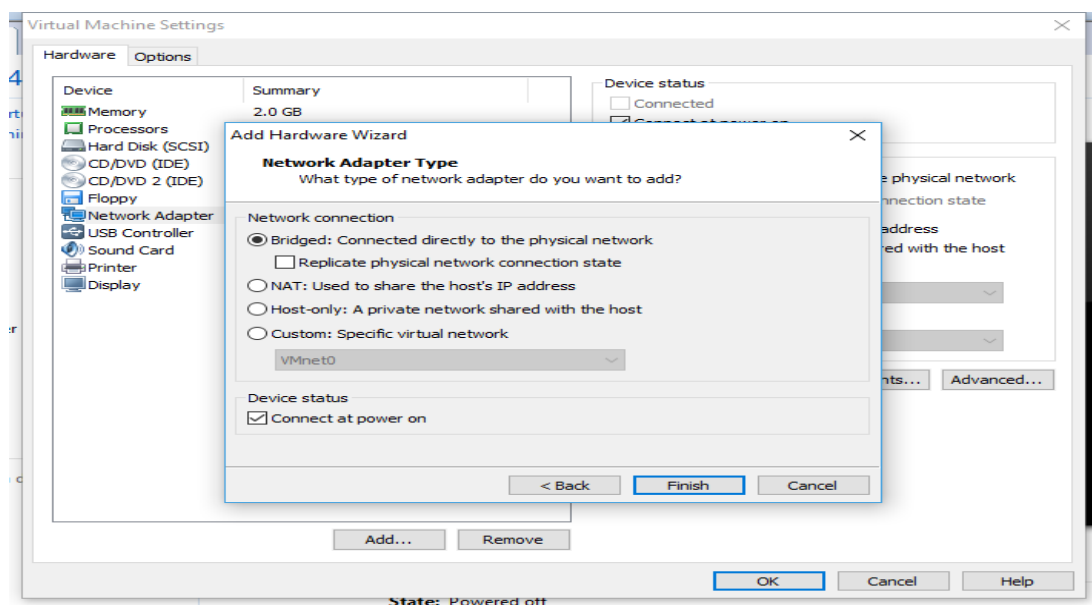


```
Open  ntp.conf /etc Save
5 # Enable this if you want statistics to be logged.
6 #statsdir /var/log/ntpstats/
7
8 statistics loopstats peerstats clockstats
9 filegen loopstats file loopstats type day enable
10 filegen peerstats file peerstats type day enable
11 filegen clockstats file clockstats type day enable
12
13 # Specify one or more NTP servers.
14
15 # Use servers from the NTP Pool Project. Approved by Ubuntu Technical Board
16 # on 2011-02-08 (LP: #104525). See http://www.pool.ntp.org/join.html for
17 # more information.
18 #pool 0.ubuntu.pool.ntp.org iburst
19 #pool 1.ubuntu.pool.ntp.org iburst
20 #pool 2.ubuntu.pool.ntp.org iburst
21 #pool 3.ubuntu.pool.ntp.org iburst
22 server 0.north-america.pool.ntp.org iburst
23 server 1.north-america.pool.ntp.org iburst
24 server 2.north-america.pool.ntp.org iburst
25 server 3.north-america.pool.ntp.org iburst
26 # Use Ubuntu's ntp server as a fallback.
27 #pool ntp.ubuntu.com
28
```

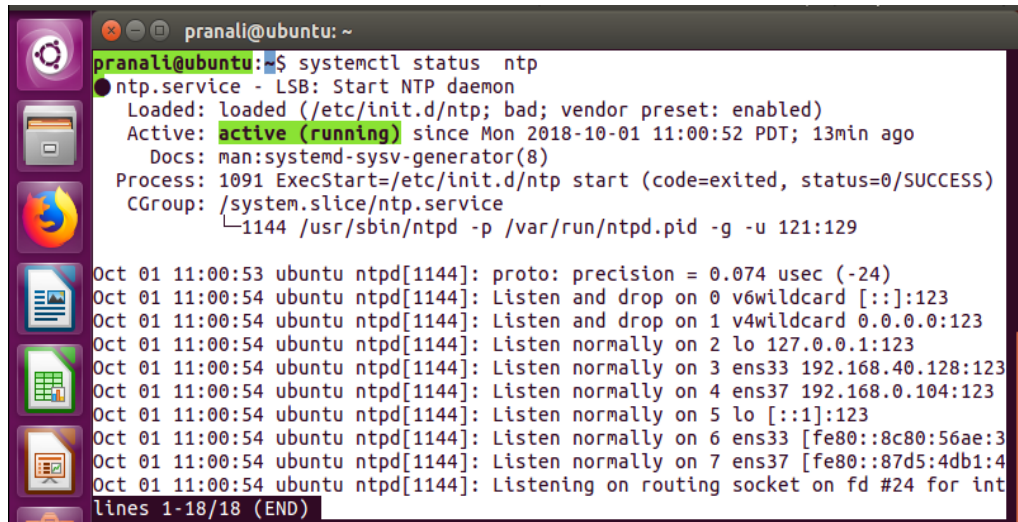
4) Systemctl restart ntp



5) Change the network adapter setting

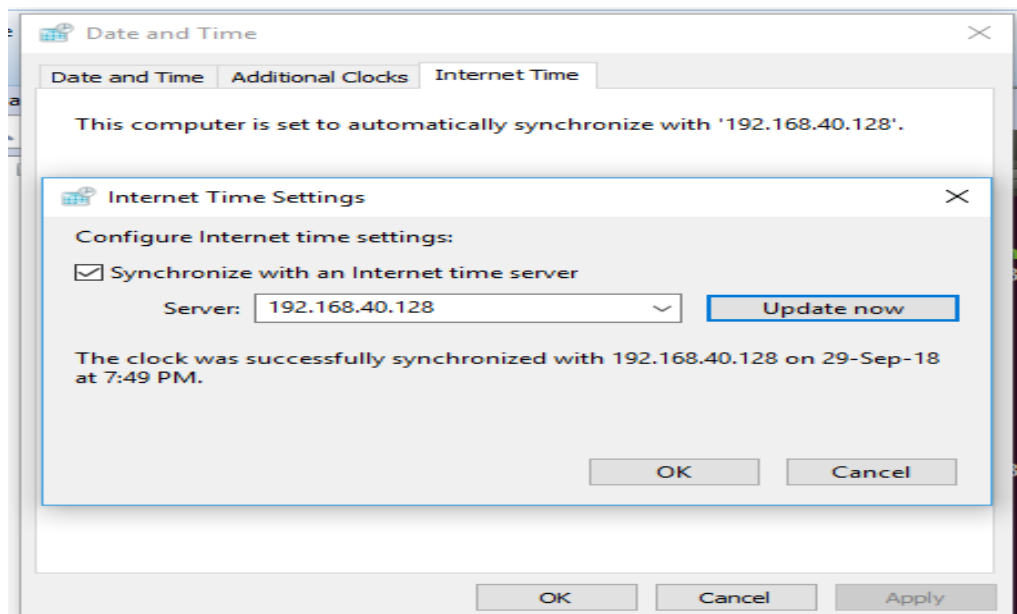


6) Systemctl status ntp



```
pranali@ubuntu: ~  
pranali@ubuntu:~$ systemctl status ntp  
● ntp.service - LSB: Start NTP daemon  
   Loaded: loaded (/etc/init.d/ntp; bad; vendor preset: enabled)  
   Active: active (running) since Mon 2018-10-01 11:00:52 PDT; 13min ago  
     Docs: man:systemd-sysv-generator(8)  
  Process: 1091 ExecStart=/etc/init.d/ntp start (code=exited, status=0/SUCCESS)  
    CGroup: /system.slice/ntp.service  
            └─1144 /usr/sbin/ntpd -p /var/run/ntpd.pid -g -u 121:129  
  
Oct 01 11:00:53 ubuntu ntpd[1144]: proto: precision = 0.074 usec (-24)  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen and drop on 0 v6wildcard [::]:123  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen and drop on 1 v4wildcard 0.0.0.0:123  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen normally on 2 lo 127.0.0.1:123  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen normally on 3 ens33 192.168.40.128:123  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen normally on 4 ens37 192.168.0.104:123  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen normally on 5 lo [::1]:123  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen normally on 6 ens33 [fe80::8c80:56ae:3  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listen normally on 7 ens37 [fe80::87d5:4db1:4  
Oct 01 11:00:54 ubuntu ntpd[1144]: Listening on routing socket on fd #24 for int  
lines 1-18/18 (END)
```

7) Synchronized the data and time.



PRACTICAL NO:-04

AIM:-

SSH Server : Password Authentication

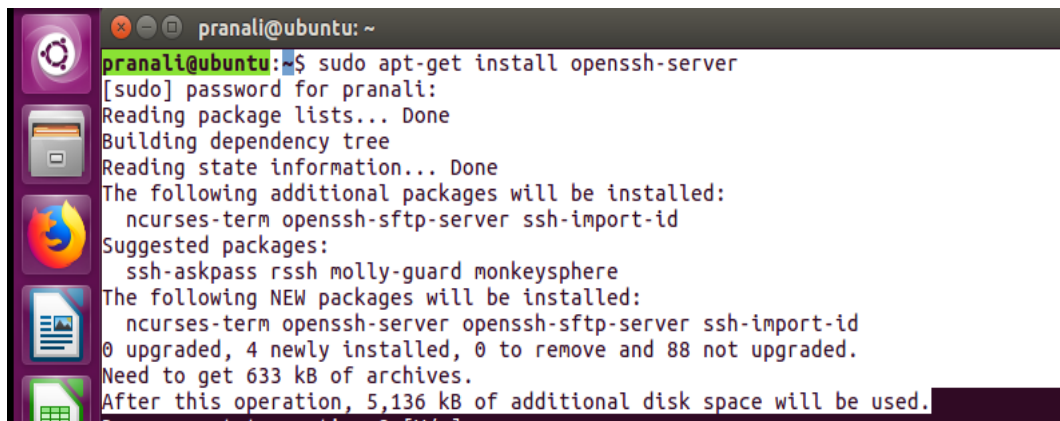
Configure SSH Server to manage a server from the remote computer, SSH Client : (Ubuntu and Windows)

COMMAND:-

Open the two ubuntu system one is client and other is server.

Run in server:

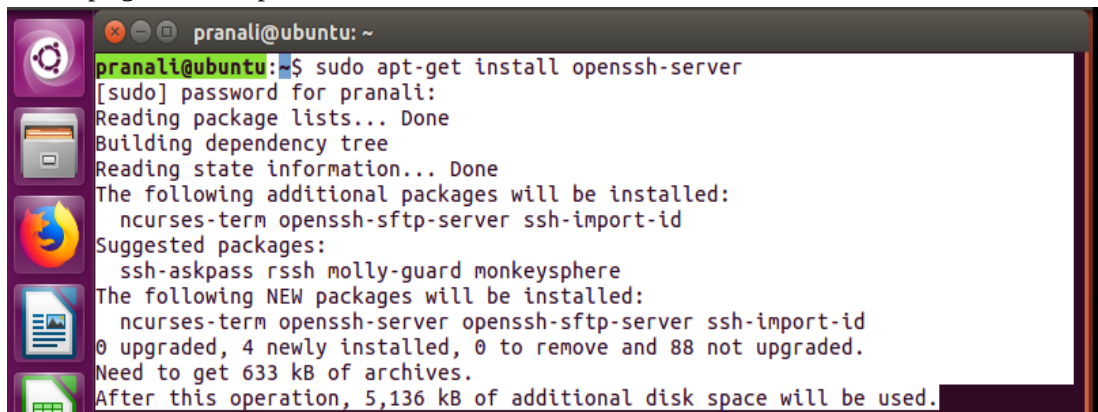
- 1) Sudo apt-get install openssh-server

A terminal window titled 'pranali@ubuntu: ~' showing the command 'sudo apt-get install openssh-server' being executed. The output includes the password prompt, package list reading, dependency tree building, and state information reading. It lists additional packages to be installed (ncurses-term, openssh-sftp-server, ssh-import-id) and suggested packages (ssh-askpass, rssh, molly-guard, monkeysphere). It also shows the number of new packages (4), the disk space required (633 kB), and the additional disk space to be used (5,136 kB).

```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install openssh-server  
[sudo] password for pranali:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  ncurses-term openssh-sftp-server ssh-import-id  
Suggested packages:  
  ssh-askpass rssh molly-guard monkeysphere  
The following NEW packages will be installed:  
  ncurses-term openssh-server openssh-sftp-server ssh-import-id  
0 upgraded, 4 newly installed, 0 to remove and 88 not upgraded.  
Need to get 633 kB of archives.  
After this operation, 5,136 kB of additional disk space will be used.
```

Run as client

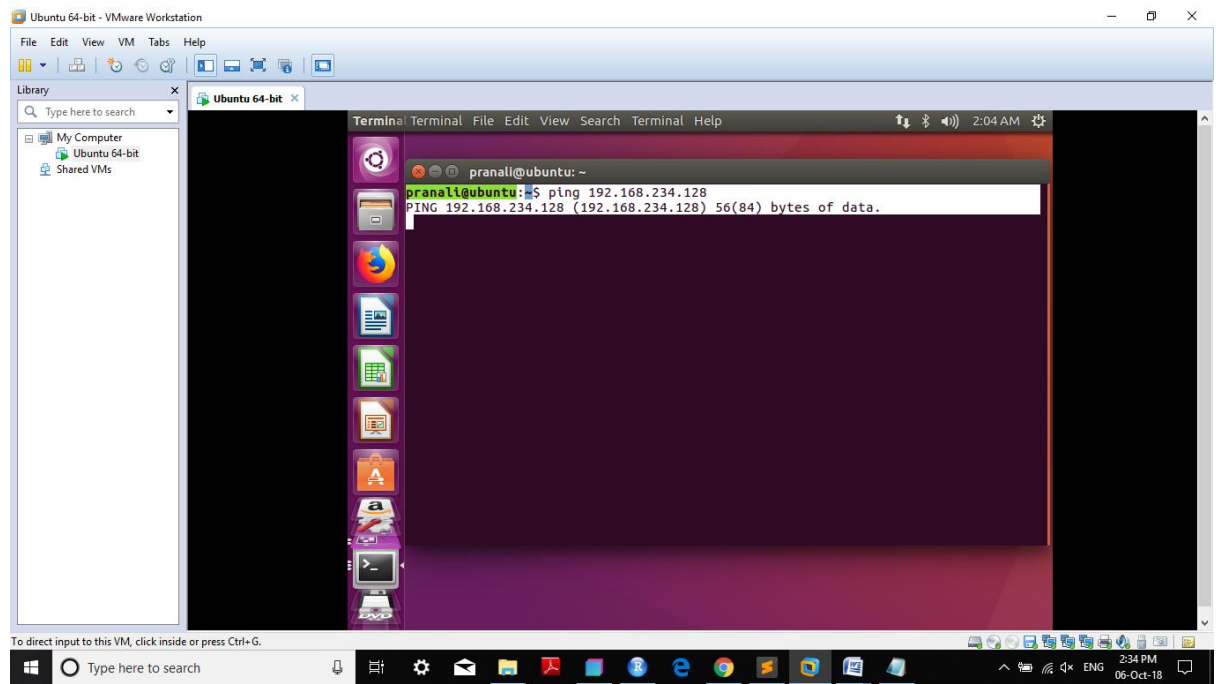
- 1) Sudo apt-get install openssh-server

A terminal window titled 'pranali@ubuntu: ~' showing the command 'sudo apt-get install openssh-server' being executed. The output is identical to the server terminal, showing the password prompt, package list reading, dependency tree building, and state information reading. It lists additional packages to be installed (ncurses-term, openssh-sftp-server, ssh-import-id) and suggested packages (ssh-askpass, rssh, molly-guard, monkeysphere). It also shows the number of new packages (4), the disk space required (633 kB), and the additional disk space to be used (5,136 kB).

```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install openssh-server  
[sudo] password for pranali:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  ncurses-term openssh-sftp-server ssh-import-id  
Suggested packages:  
  ssh-askpass rssh molly-guard monkeysphere  
The following NEW packages will be installed:  
  ncurses-term openssh-server openssh-sftp-server ssh-import-id  
0 upgraded, 4 newly installed, 0 to remove and 88 not upgraded.  
Need to get 633 kB of archives.  
After this operation, 5,136 kB of additional disk space will be used.
```

- 2) ping client ip address

LINUX SERVER ADMINISTRATION



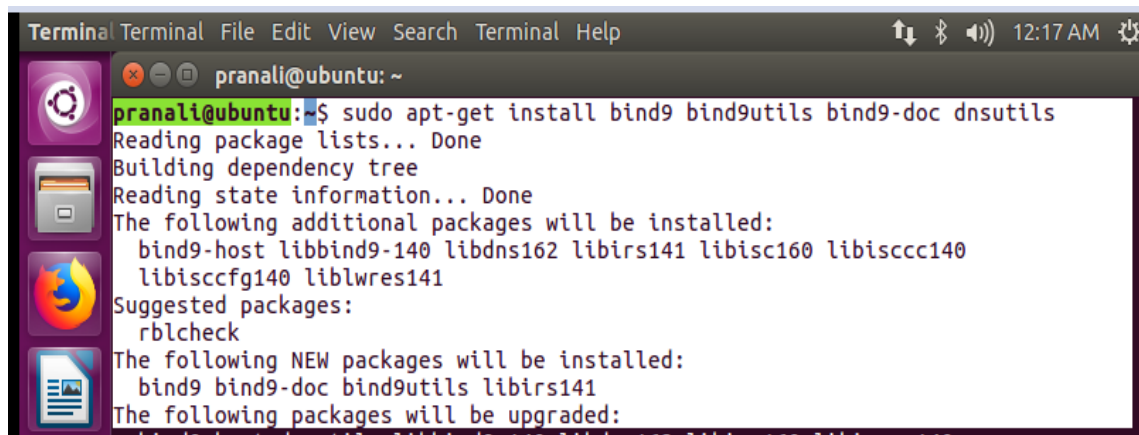
PRACTICAL NO:-05

AIM:-

Install DNS Server BIND, Configure DNS server which resolves domain name or IP address, Install BIND 9, Configure BIND.

COMMAND:-

- 1) `sudo apt-get install bind9 bind9utils bind9-doc dnsutils`



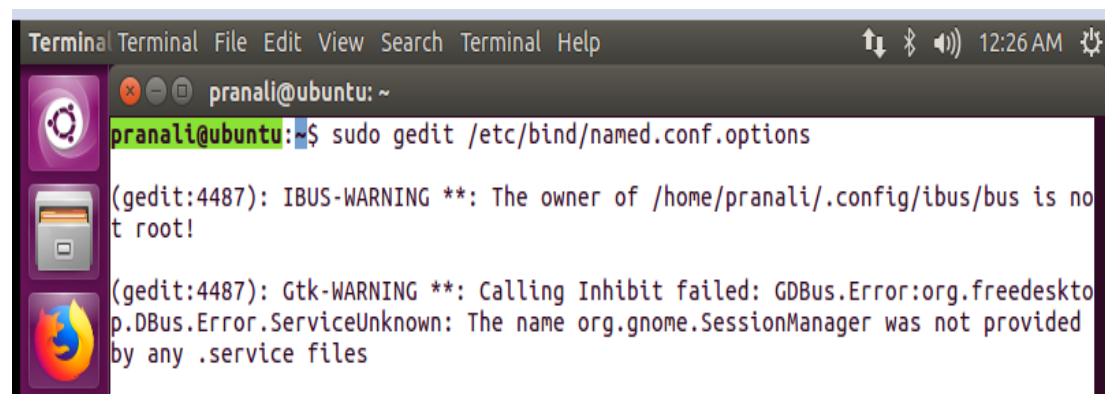
```
Terminal Terminal File Edit View Search Terminal Help 12:17 AM
pranali@ubuntu: ~
pranali@ubuntu:~$ sudo apt-get install bind9 bind9utils bind9-doc dnsutils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  bind9-host libbind9-140 libdns162 libirs141 libisc160 libisccc140
  libisccfg140 liblwres141
Suggested packages:
  rblcheck
The following NEW packages will be installed:
  bind9 bind9-doc bind9utils libirs141
The following packages will be upgraded:
```

- 2) `sudo cp /etc/bind/named.conf.options /etc/bind/named.conf.options.bak`



```
pranali@ubuntu:~$ sudo cp /etc/bind/named.conf.options /etc/bind/named.conf.options.bak
pranali@ubuntu:~$
```

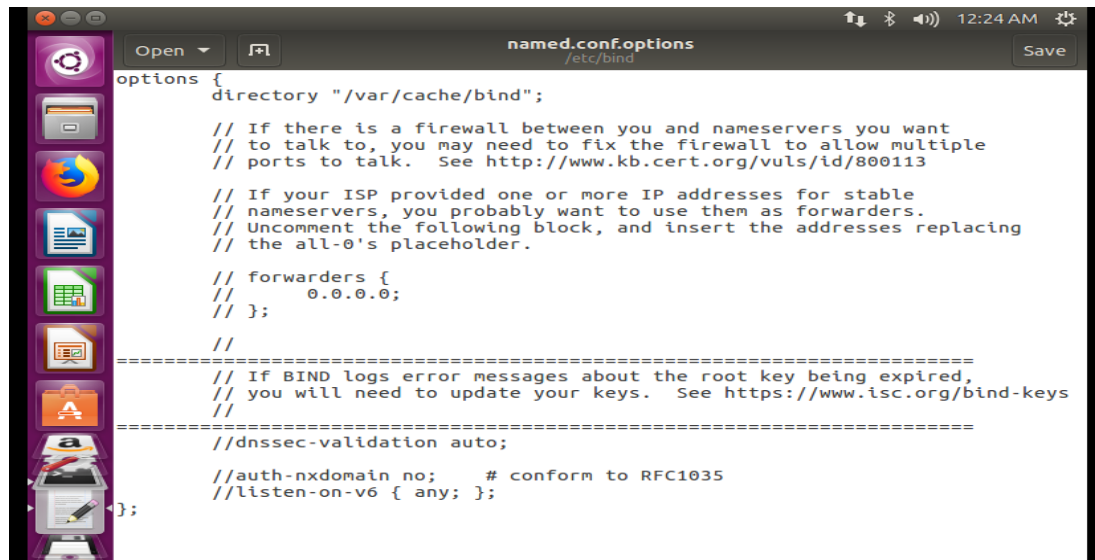
- 3) `sudo gedit /etc/bind/named.conf.options`



```
Terminal Terminal File Edit View Search Terminal Help 12:26 AM
pranali@ubuntu: ~
pranali@ubuntu:~$ sudo gedit /etc/bind/named.conf.options
(gedit:4487): IBUS-WARNING **: The owner of /home/pranali/.config/ibus/bus is not root!
(gedit:4487): Gtk-WARNING **: Calling Inhibit failed: GDBus.Error:org.freedesktop.DBus.Error.ServiceUnknown: The name org.gnome.SessionManager was not provided by any .service files
```

Comment the line number 25 to 28

LINUX SERVER ADMINISTRATION



```
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    // forwarders {
    //     0.0.0.0;
    // };

    //
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //
    //dnssec-validation auto;

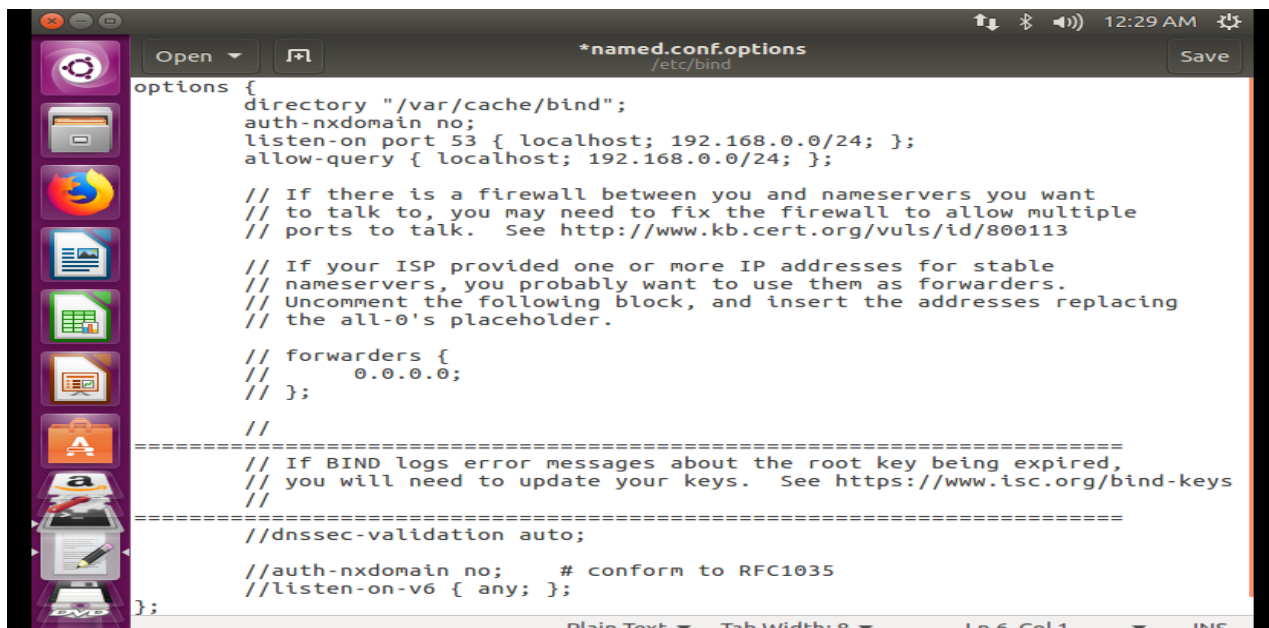
    //auth-nxdomain no;      # conform to RFC1035
    //listen-on-v6 { any; };
};
```

Write the text in line number 3

```
auth-nxdomain no;
```

```
listen-on port 53 { localhost; 192.168.0.0/24; };
```

```
allow-query { localhost; 192.168.0.0/24; };
```



```
*named.conf.options
/etc/bind
options {
    directory "/var/cache/bind";
    auth-nxdomain no;
    listen-on port 53 { localhost; 192.168.0.0/24; };
    allow-query { localhost; 192.168.0.0/24; };

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    // forwarders {
    //     0.0.0.0;
    // };

    //
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //
    //dnssec-validation auto;

    //auth-nxdomain no;      # conform to RFC1035
    //listen-on-v6 { any; };
};
```

Change

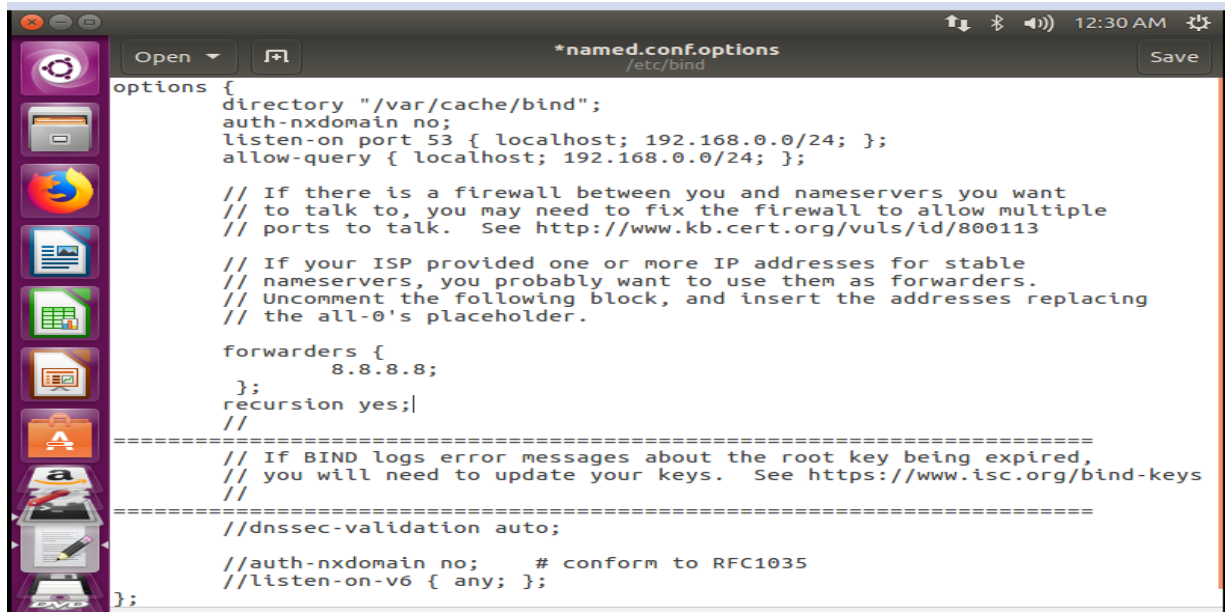
```
forwarders{
```

```
    8.8.8.8;
```

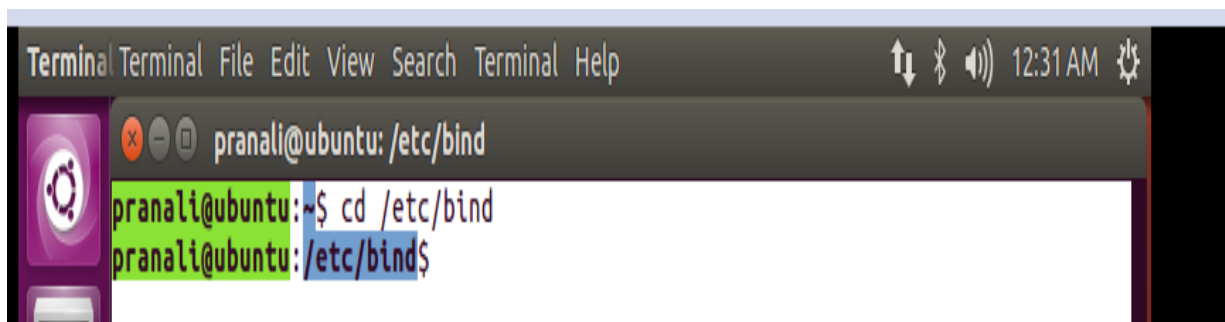
```
};
```

LINUX SERVER ADMINISTRATION

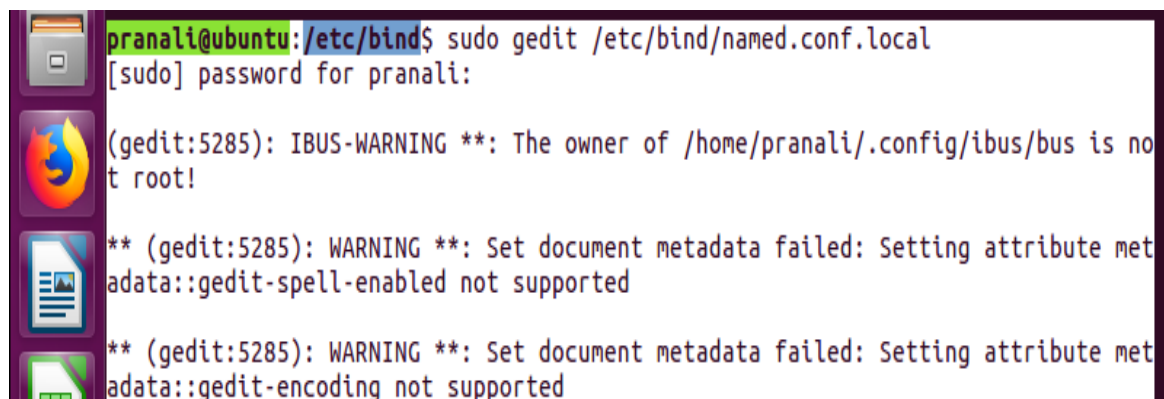
Recursion yes;



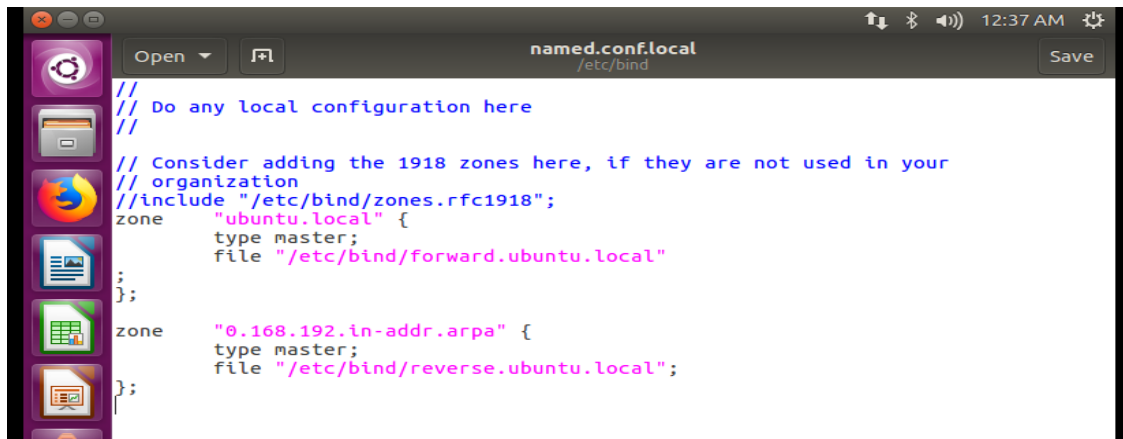
4) `cd /etc/bind`



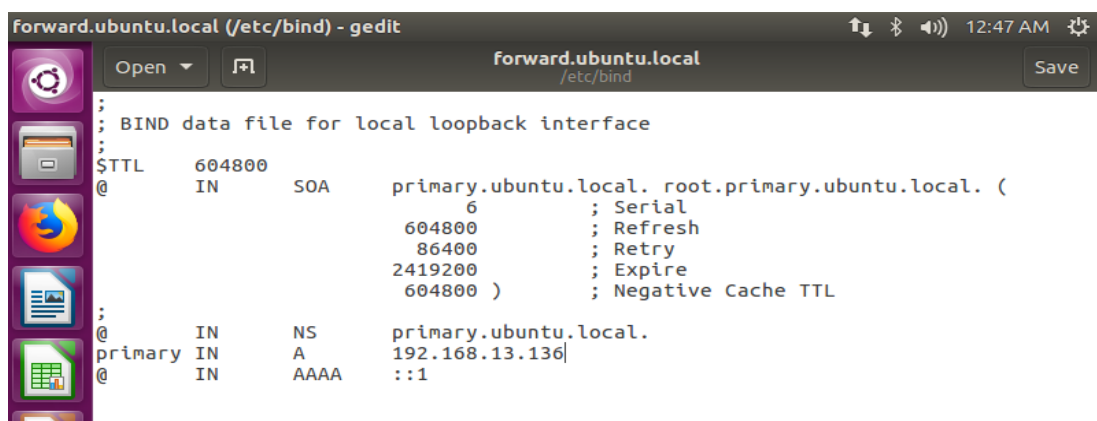
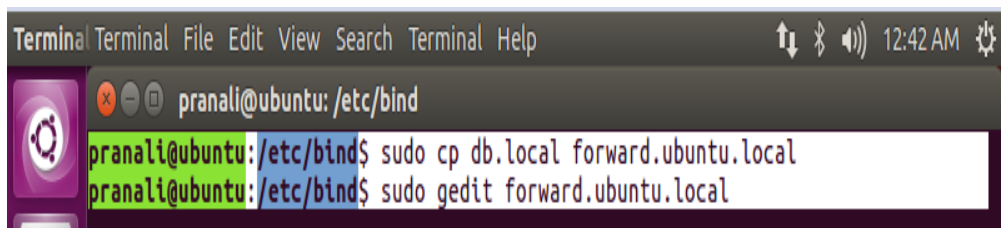
5) `sudo gedit /etc/bind/named.conf.local`



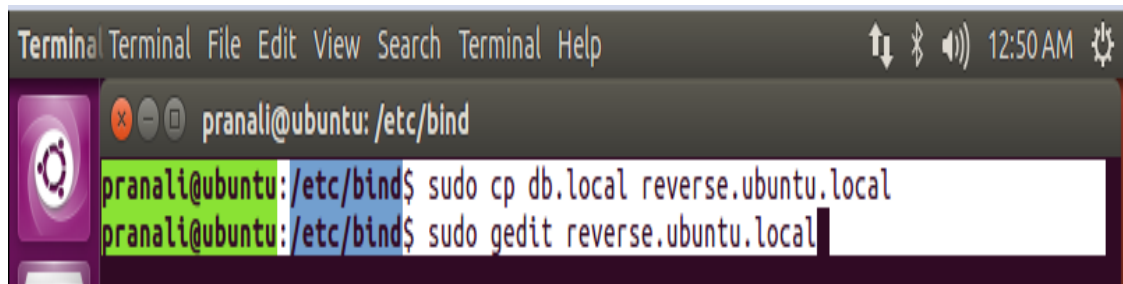
Write the following code

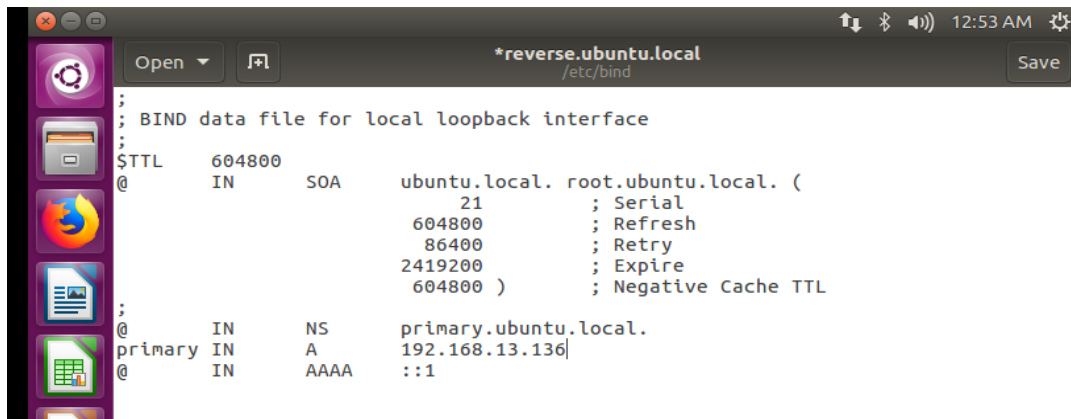


- 6) `sudo cp db.local forward.ubuntu.local`
- 7) `sudo gedit forward.ubuntu.local`



- 8) `sudo cp db.local reverse.ubuntu.local`
- 9) `sudo gedit reverse.ubuntu.local`



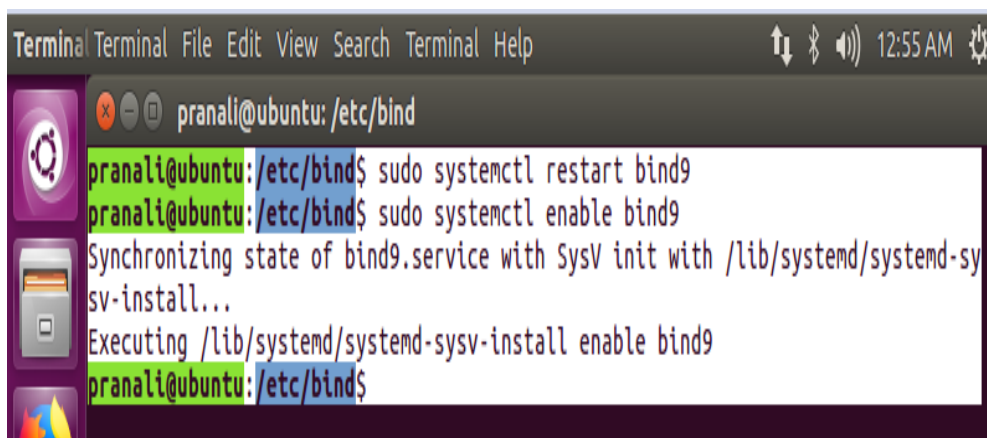


```
*reverse.ubuntu.local
/etc/bind

;
; BIND data file for local loopback interface
;
$TTL      604800
@         IN      SOA      ubuntu.local. root.ubuntu.local. (
        21      ; Serial
        604800   ; Refresh
        86400    ; Retry
        2419200  ; Expire
        604800   ; Negative Cache TTL
);
;
@         IN      NS       primary.ubuntu.local.
primary   IN      A        192.168.13.136
@         IN      AAAA     ::1
```

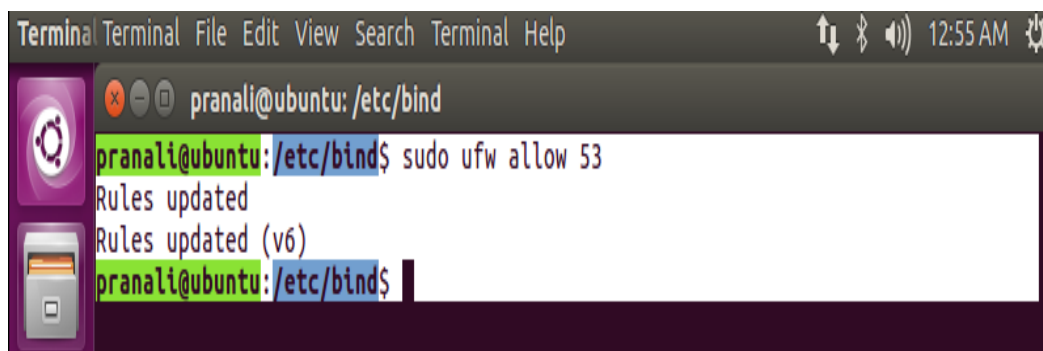
10) sudo systemctl restart bind9

11) sudo systemctl enable bind9



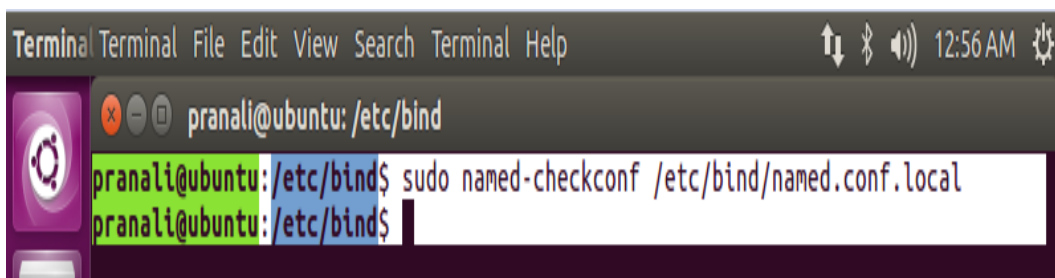
```
pranali@ubuntu: /etc/bind$ sudo systemctl restart bind9
pranali@ubuntu: /etc/bind$ sudo systemctl enable bind9
Synchronizing state of bind9.service with SysV init with /lib/systemd/systemd-sy
sv-install...
Executing /lib/systemd/systemd-sysv-install enable bind9
pranali@ubuntu: /etc/bind$
```

12) sudo ufw allow 53



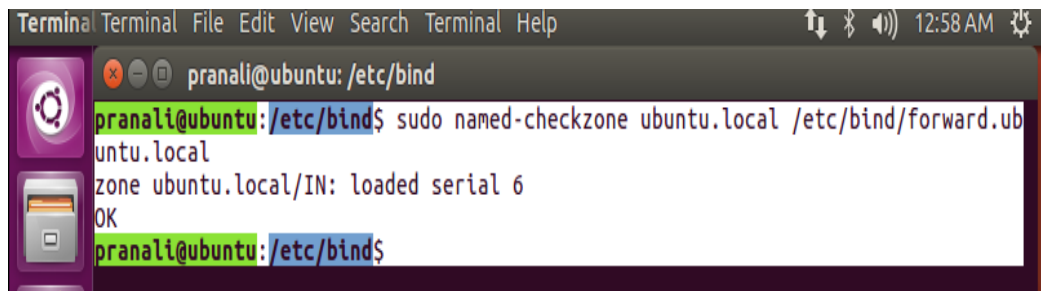
```
pranali@ubuntu: /etc/bind$ sudo ufw allow 53
Rules updated
Rules updated (v6)
pranali@ubuntu: /etc/bind$
```

13) sudo named-checkconf /etc/bind/named.conf.local



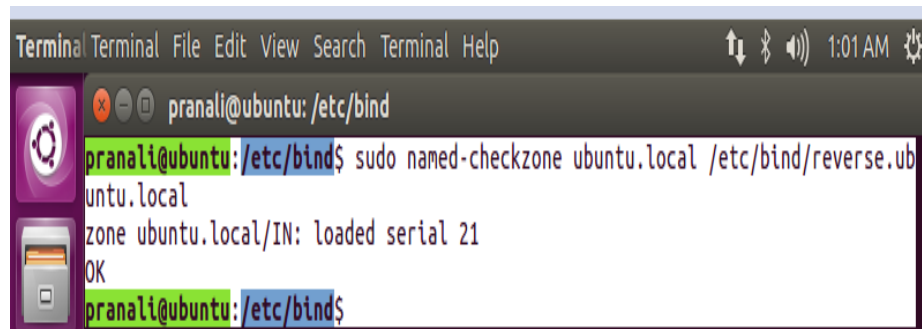
```
pranali@ubuntu: /etc/bind$ sudo named-checkconf /etc/bind/named.conf.local
pranali@ubuntu: /etc/bind$
```

14) `sudo named -checkzone ubuntu.local /etc/bind/forward.ubuntu.local`



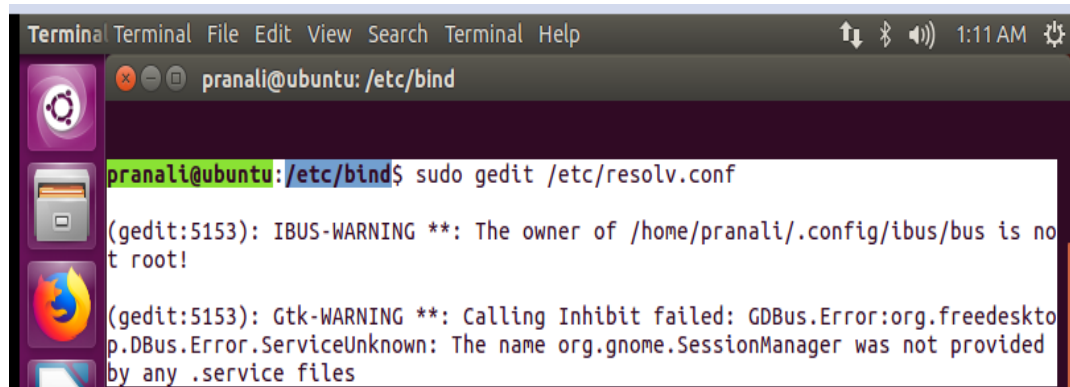
```
Terminal Terminal File Edit View Search Terminal Help
pranali@ubuntu: /etc/bind
pranali@ubuntu: /etc/bind$ sudo named-checkzone ubuntu.local /etc/bind/forward.ubuntu.local
zone ubuntu.local/IN: loaded serial 6
OK
pranali@ubuntu: /etc/bind$
```

15) `sudo named -checkzone ubuntu.local /etc/bind/reverse.ubuntu.local`



```
Terminal Terminal File Edit View Search Terminal Help
pranali@ubuntu: /etc/bind
pranali@ubuntu: /etc/bind$ sudo named-checkzone ubuntu.local /etc/bind/reverse.ubuntu.local
zone ubuntu.local/IN: loaded serial 21
OK
pranali@ubuntu: /etc/bind$
```

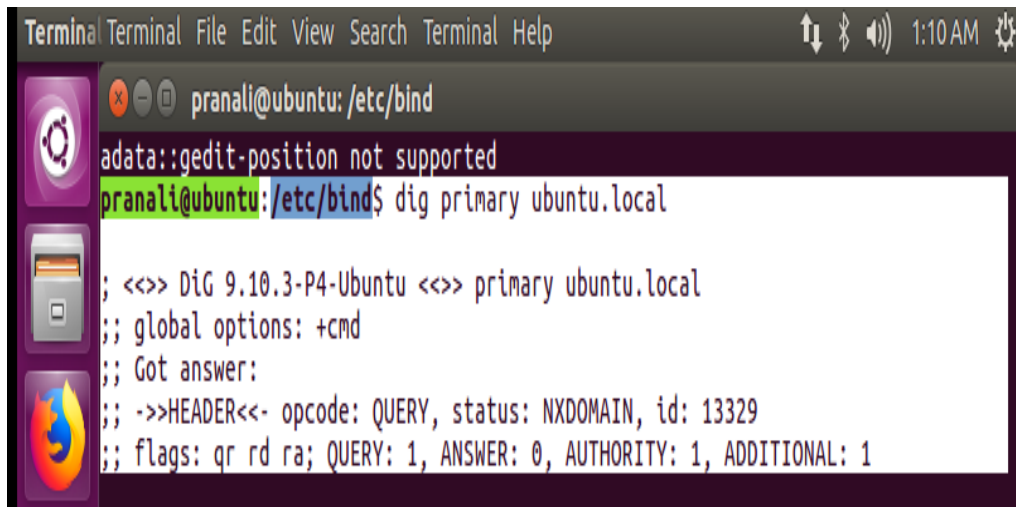
16) `sudo gedit /etc/resolv.conf`



```
Terminal Terminal File Edit View Search Terminal Help
pranali@ubuntu: /etc/bind
pranali@ubuntu: /etc/bind$ sudo gedit /etc/resolv.conf
(gedit:5153): IBUS-WARNING **: The owner of /home/pranali/.config/ibus/bus is not root!
(gedit:5153): Gtk-WARNING **: Calling Inhibit failed: GDBus.Error:org.freedesktop.DBus.Error.ServiceUnknown: The name org.gnome.SessionManager was not provided by any .service files
```



17) dig primary ubuntu.local



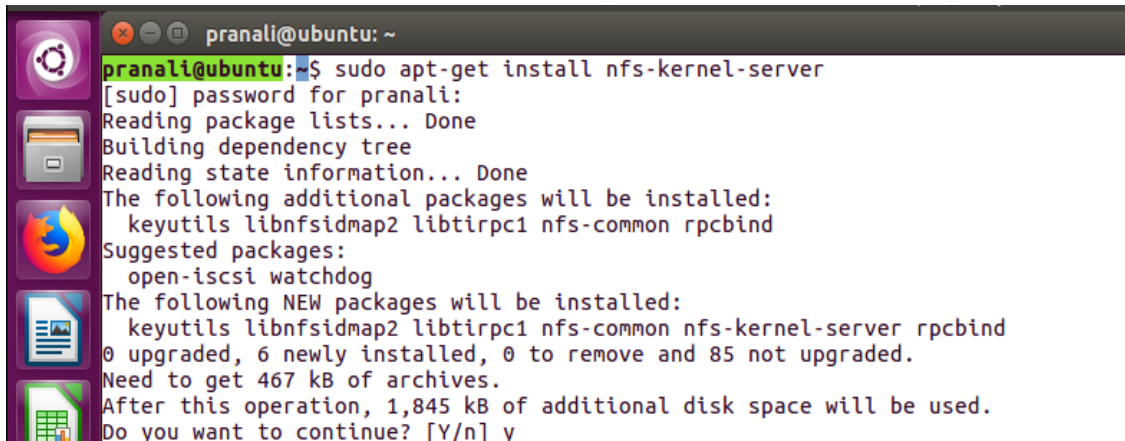
PRACTICAL NO:-06

AIM:-

Configure NFS Server to share directories on your Network, Configure NFS Client.
(Ubuntu and Windows Client OS)

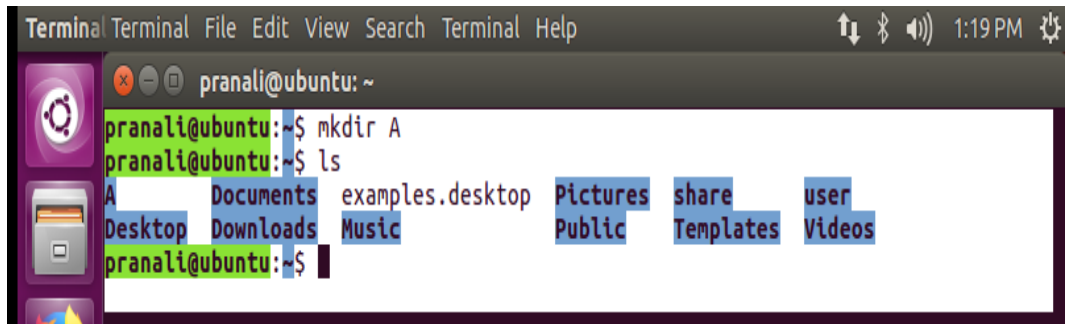
COMMAND:-

- 1) Sudo apt-get install nfs-kernel-server



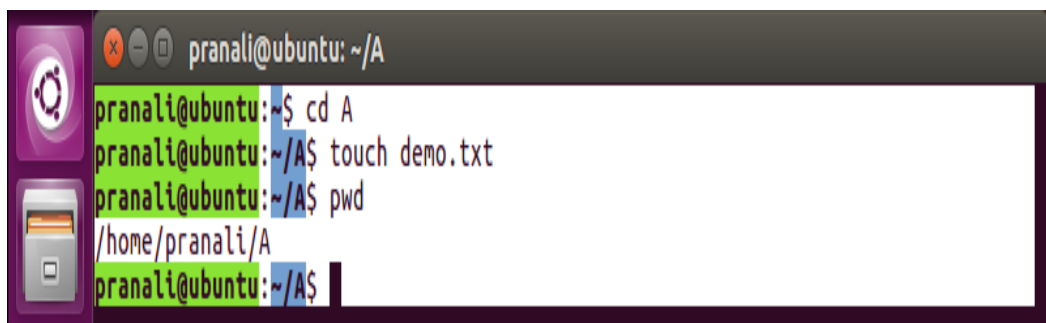
```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install nfs-kernel-server  
[sudo] password for pranali:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  keyutils libnfsidmap2 libtirpc1 nfs-common rpcbind  
Suggested packages:  
  open-iscsi watchdog  
The following NEW packages will be installed:  
  keyutils libnfsidmap2 libtirpc1 nfs-common nfs-kernel-server rpcbind  
0 upgraded, 6 newly installed, 0 to remove and 85 not upgraded.  
Need to get 467 kB of archives.  
After this operation, 1,845 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y
```

- 2) Mkdir A
- 3) Ls



```
Terminal Terminal File Edit View Search Terminal Help  
pranali@ubuntu: ~  
pranali@ubuntu:~$ mkdir A  
pranali@ubuntu:~$ ls  
A Desktop Documents examples.desktop Pictures share user  
Downloads Music Public Templates Videos  
pranali@ubuntu:~$
```

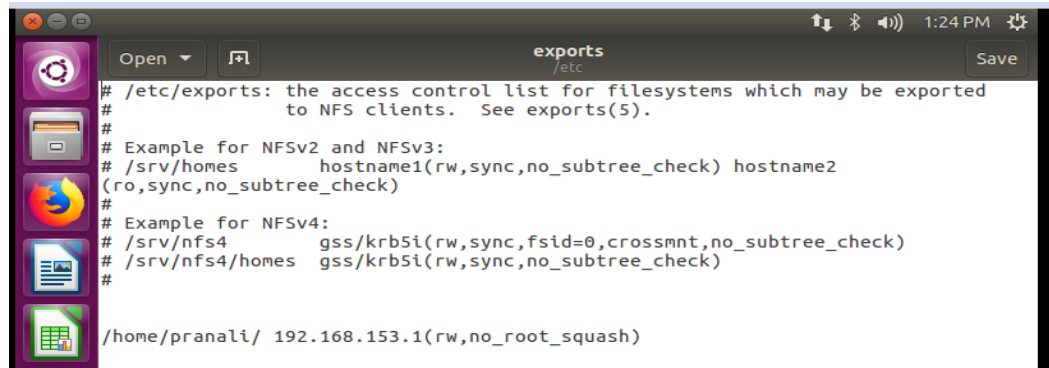
- 4) Cd A
- 5) Touch demo.txt
- 6) Pwd



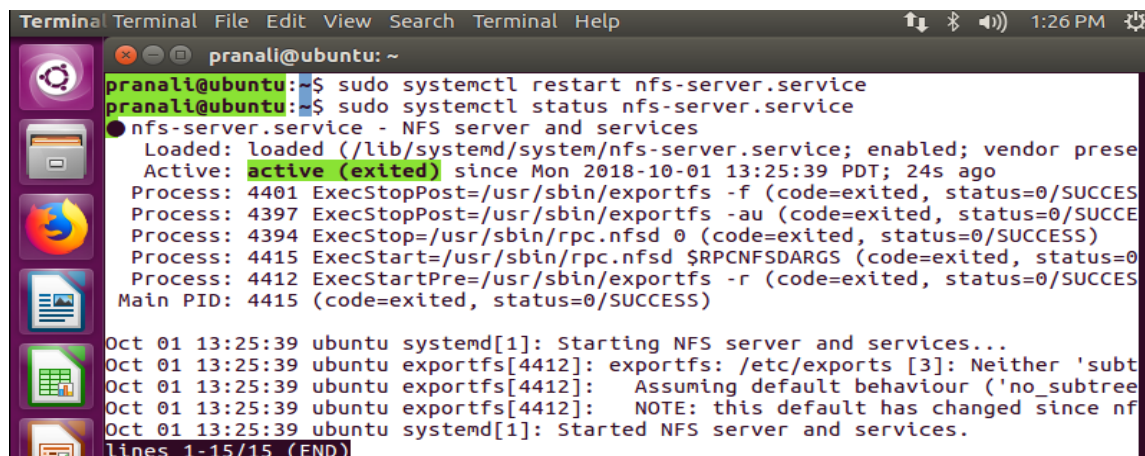
```
pranali@ubuntu: ~/A  
pranali@ubuntu:~$ cd A  
pranali@ubuntu:~/A$ touch demo.txt  
pranali@ubuntu:~/A$ pwd  
/home/pranali/A  
pranali@ubuntu:~/A$
```

LINUX SERVER ADMINISTRATION

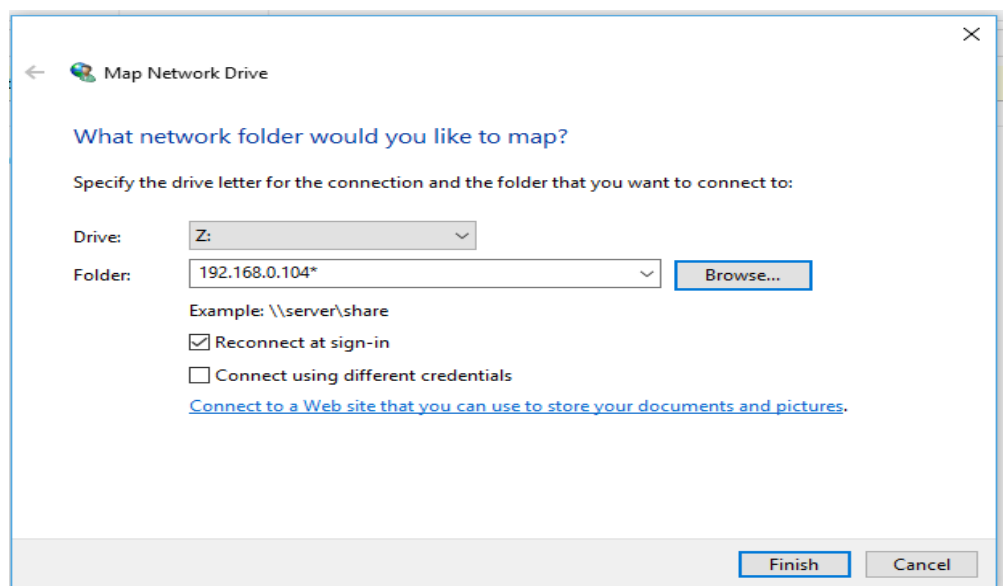
- 7) Sudo gedit /etc/exports
/home/Pranali/ ip address(rw,no_root_squash)



- 8) Sudo systemctl restart nfs-server.service
9) Sudo systemctl status nfs-server.service



- 10) My computer > network>map network drive> ip address

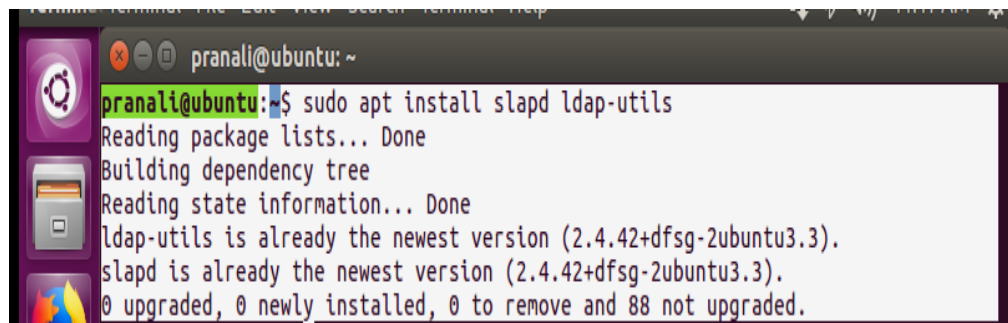


PRACTICAL NO:-07**AIM:-**

Configure LDAP Server, Configure LDAP Server in order to share users' accounts in your local networks, Add LDAP User Accounts in the OpenLDAP Server, Configure LDAP Client in order to share users' accounts in your local networks. Install phpLDAPadmin to operate LDAP server via Web browser.

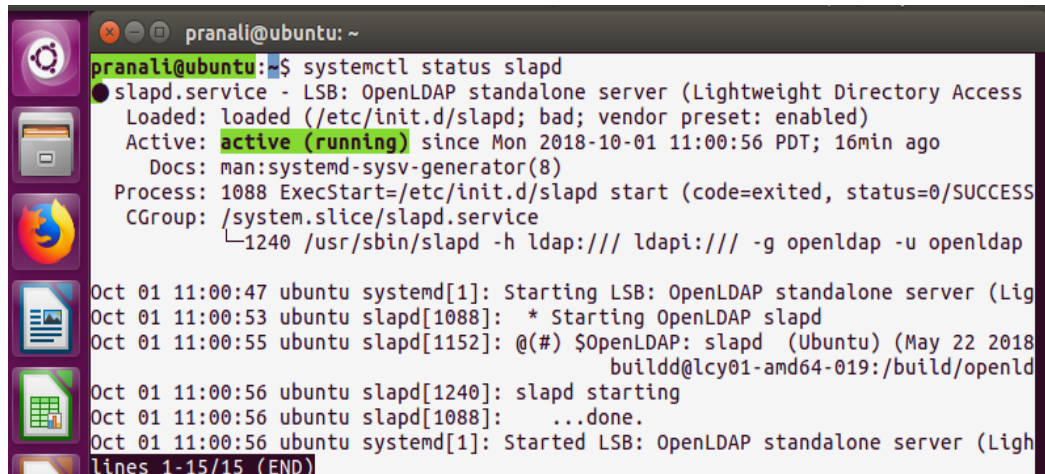
COMMAND:-

- 1) Sudo apt install slapd ldap-utils



```
pranali@ubuntu:~$ sudo apt install slapd ldap-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
ldap-utils is already the newest version (2.4.42+dfsg-2ubuntu3.3).
slapd is already the newest version (2.4.42+dfsg-2ubuntu3.3).
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
```

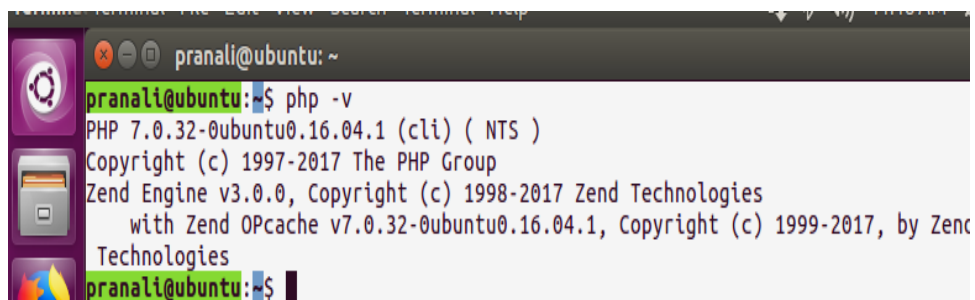
- 2) Systemctl status slapd



```
pranali@ubuntu:~$ systemctl status slapd
● slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access
   Loaded: loaded (/etc/init.d/slapd; bad; vendor preset: enabled)
   Active: active (running) since Mon 2018-10-01 11:00:56 PDT; 16min ago
     Docs: man:systemd-sysv-generator(8)
   Process: 1088 ExecStart=/etc/init.d/slapd start (code=exited, status=0/SUCCESS)
    CGroup: /system.slice/slapd.service
            └─1240 /usr/sbin/slapd -h ldap:/// ldapi:/// -g openldap -u openldap

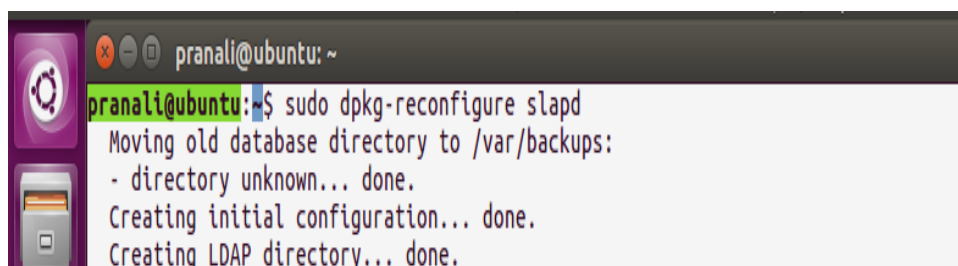
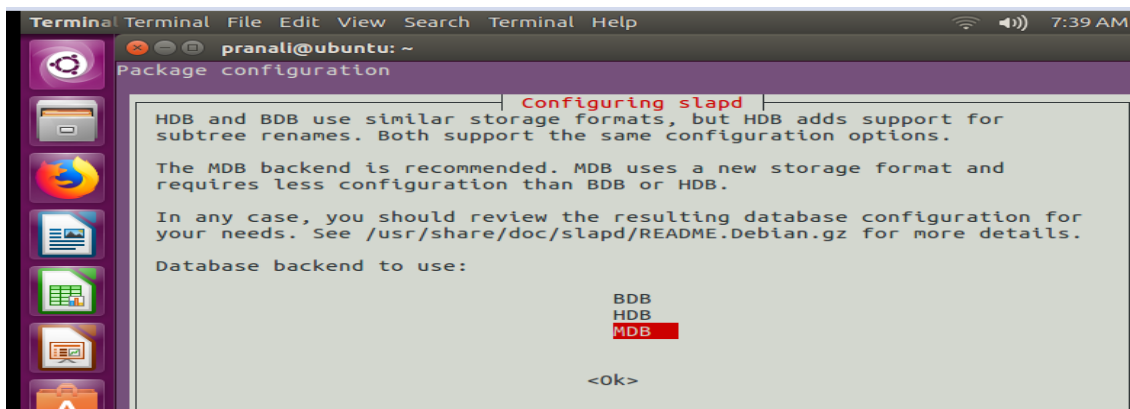
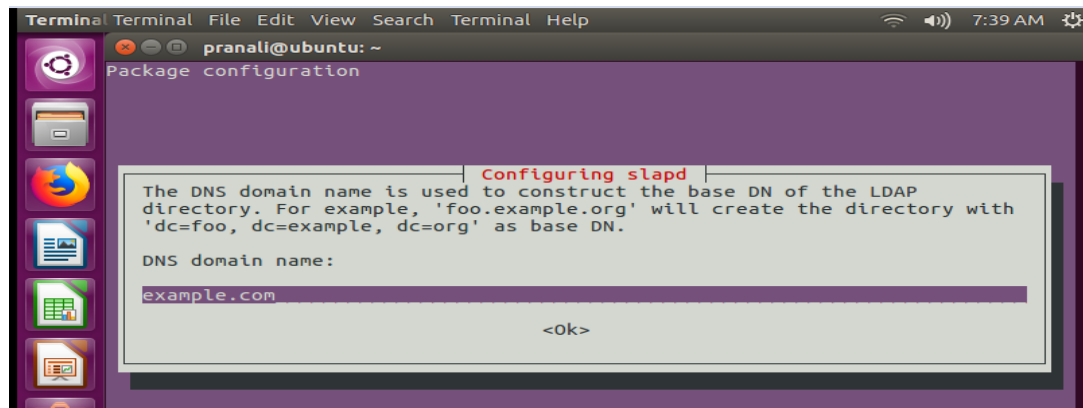
Oct 01 11:00:47 ubuntu systemd[1]: Starting LSB: OpenLDAP standalone server (Lig
Oct 01 11:00:53 ubuntu slapd[1088]: * Starting OpenLDAP slapd
Oct 01 11:00:55 ubuntu slapd[1152]: @(#) $OpenLDAP: slapd (Ubuntu) (May 22 2018
                        build@lcy01-amd64-019:/build/openld
Oct 01 11:00:56 ubuntu slapd[1240]: slapd starting
Oct 01 11:00:56 ubuntu slapd[1088]: ...done.
Oct 01 11:00:56 ubuntu systemd[1]: Started LSB: OpenLDAP standalone server (Ligh
lines 1-15/15 (END)
```

- 3) Php -v



```
pranali@ubuntu:~$ php -v
PHP 7.0.32-0ubuntu0.16.04.1 (cli) ( NTS )
Copyright (c) 1997-2017 The PHP Group
Zend Engine v3.0.0, Copyright (c) 1998-2017 Zend Technologies
    with Zend OPcache v7.0.32-0ubuntu0.16.04.1, Copyright (c) 1999-2017, by Zend
Technologies
pranali@ubuntu:~$
```

4) Sudo dpkg-reconfigure slapd

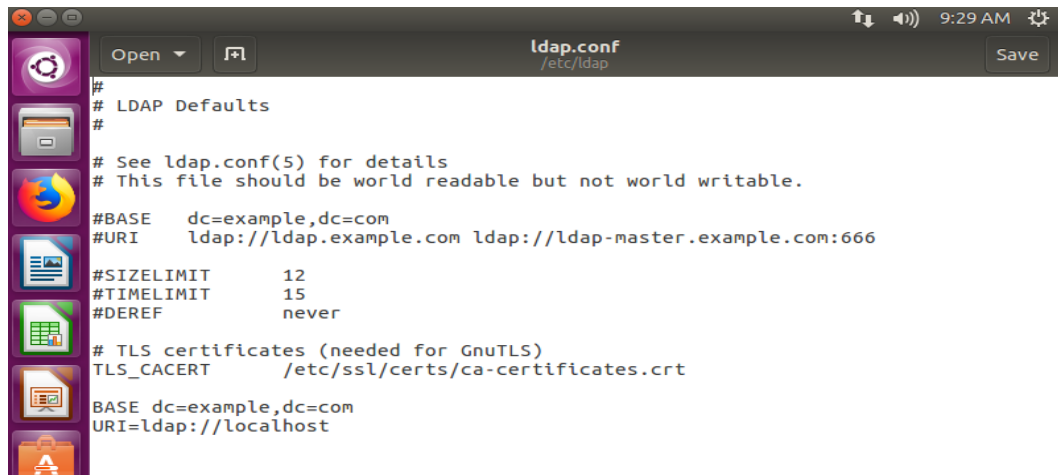


5) Sudo gedit /etc/ldap/ldap.conf

Write the following in ldap.conf file:-

BASE dc=example,dc=com

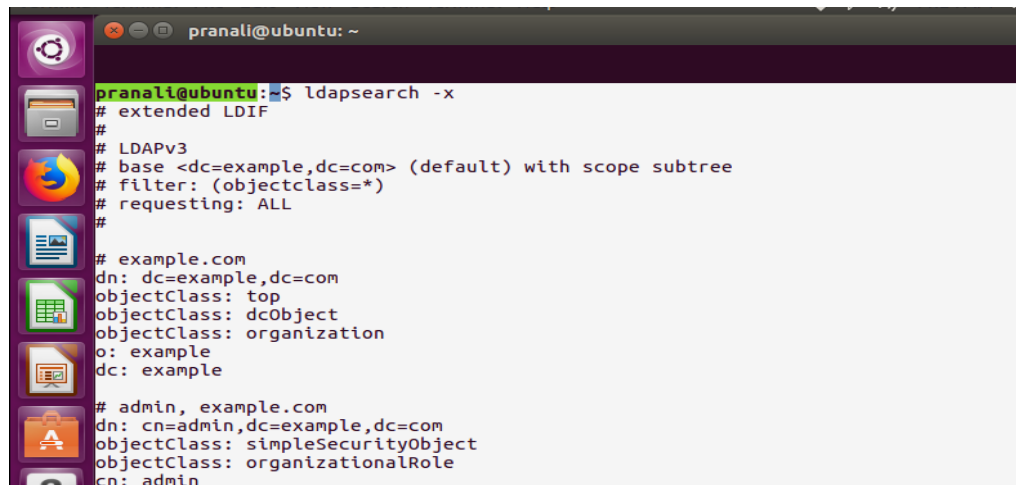
URI=ldap: //localhost



A screenshot of a text editor window showing the contents of the `/etc/ldap/ldap.conf` file. The file contains LDAP configuration settings, including defaults, base URI, size and time limits, and TLS certificates. The configuration is as follows:

```
#  
# LDAP Defaults  
#  
# See ldap.conf(5) for details  
# This file should be world readable but not world writable.  
#BASE    dc=example,dc=com  
#URI      ldap://ldap.example.com ldap://ldap-master.example.com:666  
  
#SIZELIMIT    12  
#TIMELIMIT    15  
#DEREF        never  
  
# TLS certificates (needed for GnuTLS)  
TLS_CACERT    /etc/ssl/certs/ca-certificates.crt  
  
BASE dc=example,dc=com  
URI=ldap://localhost
```

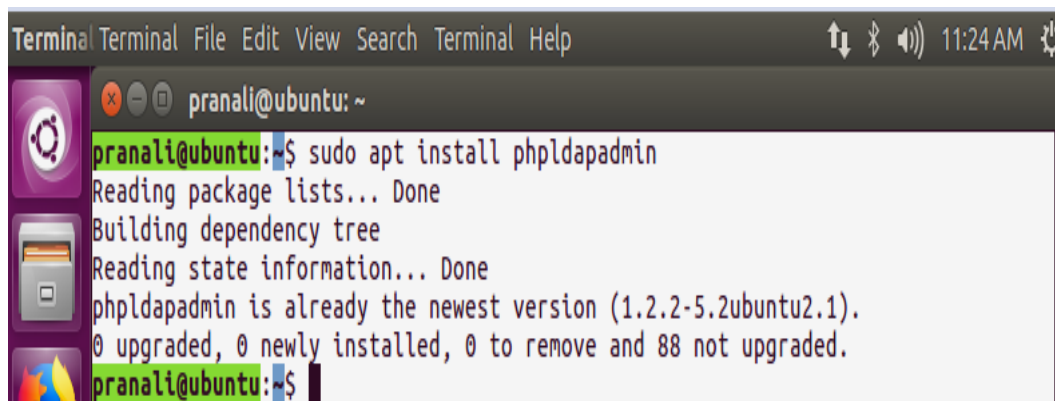
6) `ldapsearch -x`



A screenshot of a terminal window showing the output of the `ldapsearch -x` command. The output displays LDAP search details, including the base, filter, and requested attributes, followed by the search results for the `example.com` domain and the `admin` user.

```
pranali@ubuntu: ~  
pranali@ubuntu:~$ ldapsearch -x  
# extended LDIF  
#  
# LDAPv3  
# base <dc=example,dc=com> (default) with scope subtree  
# filter: (objectclass=*)  
# requesting: ALL  
#  
# example.com  
dn: dc=example,dc=com  
objectClass: top  
objectClass: dcObject  
objectClass: organization  
o: example  
dc: example  
  
# admin, example.com  
dn: cn=admin,dc=example,dc=com  
objectClass: simpleSecurityObject  
objectClass: organizationalRole  
cn: admin
```

7) `sudo apt install phpldapadmin`



A screenshot of a terminal window showing the output of the `sudo apt install phpldapadmin` command. The output indicates that the package is already installed and is the newest version available.

```
Terminal Terminal File Edit View Search Terminal Help  
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt install phpldapadmin  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
phpldapadmin is already the newest version (1.2.2-5.2ubuntu2.1).  
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.  
pranali@ubuntu:~$
```

8) systemctl status slapd

```
pranali@ubuntu: ~$ systemctl status slapd
● slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access
   Loaded: loaded (/etc/init.d/slapd; bad; vendor preset: enabled)
   Active: active (running) since Mon 2018-10-01 11:19:33 PDT; 5min ago
     Docs: man:systemd-sysv-generator(8)
  Process: 3515 ExecStart=/etc/init.d/slapd start (code=exited, status=0/SUCCESS)
    CGroup: /system.slice/slapd.service
            └─3522 /usr/sbin/slapd -h ldap:/// ldapi:/// -g openldap -u openldap

Oct 01 11:19:32 ubuntu systemd[1]: Starting LSB: OpenLDAP standalone server (Lig
Oct 01 11:19:33 ubuntu slapd[3515]: * Starting OpenLDAP slapd
Oct 01 11:19:33 ubuntu slapd[3521]: @(#) $OpenLDAP: slapd (Ubuntu) (May 22 2018
        build@lcy01-amd64-019:/build/openld
Oct 01 11:19:33 ubuntu slapd[3522]: slapd starting
Oct 01 11:19:33 ubuntu slapd[3515]: ...done.
Oct 01 11:19:33 ubuntu systemd[1]: Started LSB: OpenLDAP standalone server (Ligh
lines 1-15/15 (END)
```

9) sudo nano /etc/phpldapadmin/config.php

delete the array value in line number 300 using(shift+ctrl+-)key

```
GNU nano 2.5.3 File: /etc/phpldapadmin/config.php Modified

'ldapi://%2fusr%2flocal%2fvar%2frun%2fldapi'
(Unix socket at /usr/local/var/run/ldap) */
$servers->setValue('server','host','127.0.0.1');

/* The port your LDAP server listens on (no quotes). 389 is standard. */
// $servers->setValue('server','port',389);

/* Array of base DNS of your LDAP server. Leave this blank to have phpLDAPadmin
auto-detect it for you. */
$servers->setValue('server','base',array());

/* Five options for auth_type:
1. 'cookie': you will login via a web form, and a client-side cookie will
store your login dn and password.
2. 'session': same as cookie but your login dn and password are stored on the
web server in a persistent session variable.
3. 'http': same as session but your login dn and password are retrieved via
HTTP authentication.
4. 'config': specify your login dn and password here in this config file. No

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line
```

Remove the comment in line number 161



```
pranali@ubuntu: ~
GNU nano 2.5.3      File: /etc/phpldapadmin/config.php      Modified

# $config->custom->appearance['tree'] = 'HTMLTree';
/* Just show your custom templates. */
// $config->custom->appearance['custom_templates_only'] = false;

/* Disable the default template. */
// $config->custom->appearance['disable_default_template'] = false;

/* Hide the warnings for invalid objectClasses/attributes in templates. */
$config->custom->appearance['hide_template_warning'] = false;

/* Set to true if you would like to hide header and footer parts. */
// $config->custom->appearance['minimalMode'] = false;

/* Configure what objects are shown in left hand tree */
// $config->custom->appearance['tree_filter'] = '(objectclass=*)';

/* The height and width of the tree. If these values are not set, then
no tree scroll bars are provided. */
```

- 10) Open the browser and refresh it.
Localhost/phpldapadmin/

PRACTICAL NO:-08

AIM:-

Configure NIS Server in order to share users' accounts in your local networks,
Configure NIS Client to bind NIS Server.

COMMAND:-

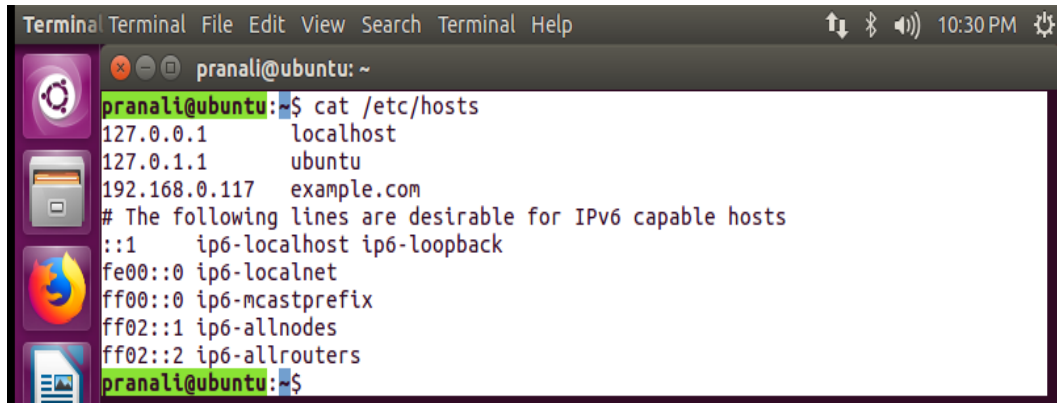
- 1) Change network adapter setting into Bridged.
- 2) Check the ip address of system

```
pranali@ubuntu: ~  
pranali@ubuntu:~$ ifconfig  
ens33      Link encap:Ethernet  HWaddr 00:0c:29:07:fb:9d  
            inet addr:192.168.13.136  Bcast:192.168.13.255  Mask:255.255.255.0  
            inet6 addr: fe80::6824:26a8:f3ef:5f5f/64 Scope:Link  
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1  
            RX packets:287 errors:0 dropped:0 overruns:0 frame:0  
            TX packets:195 errors:0 dropped:0 overruns:0 carrier:0  
            collisions:0 txqueuelen:1000  
            RX bytes:36084 (36.0 KB)  TX bytes:26027 (26.0 KB)
```

- 3) sudo nano /etc/hosts

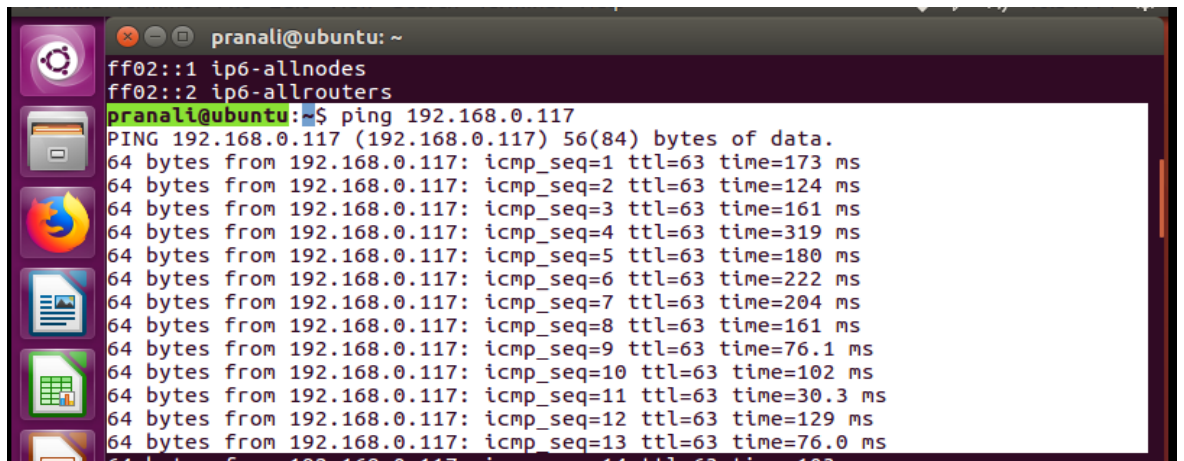
```
pranali@ubuntu: ~  
GNU nano 2.5.3      File: /etc/hosts      Modified  
127.0.0.1          localhost  
127.0.1.1          ubuntu  
192.168.0.117      example.com  
# The following lines are desirable for IPv6 capable hosts  
::1               ip6-localhost ip6-loopback  
fe00::0           ip6-localnet  
ff00::0           ip6-mcastprefix  
ff02::1           ip6-allnodes  
ff02::2           ip6-allrouters
```

4) cat /etc/hosts



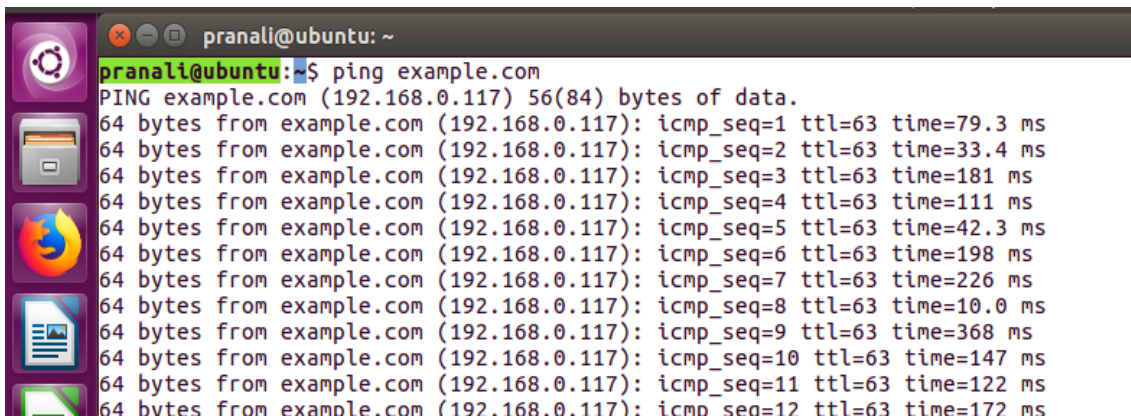
```
pranali@ubuntu: ~  
pranali@ubuntu:~$ cat /etc/hosts  
127.0.0.1 localhost  
127.0.1.1 ubuntu  
192.168.0.117 example.com  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
pranali@ubuntu:~$
```

5) ping 192.168.0.117



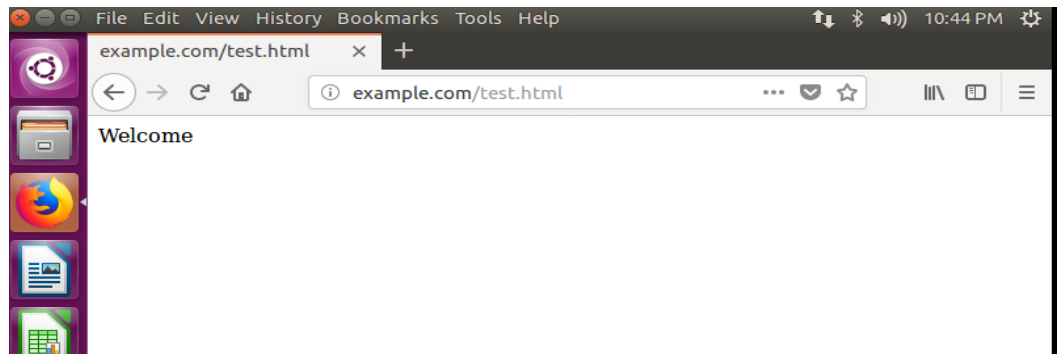
```
pranali@ubuntu: ~  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
pranali@ubuntu:~$ ping 192.168.0.117  
PING 192.168.0.117 (192.168.0.117) 56(84) bytes of data.  
64 bytes from 192.168.0.117: icmp_seq=1 ttl=63 time=173 ms  
64 bytes from 192.168.0.117: icmp_seq=2 ttl=63 time=124 ms  
64 bytes from 192.168.0.117: icmp_seq=3 ttl=63 time=161 ms  
64 bytes from 192.168.0.117: icmp_seq=4 ttl=63 time=319 ms  
64 bytes from 192.168.0.117: icmp_seq=5 ttl=63 time=180 ms  
64 bytes from 192.168.0.117: icmp_seq=6 ttl=63 time=222 ms  
64 bytes from 192.168.0.117: icmp_seq=7 ttl=63 time=204 ms  
64 bytes from 192.168.0.117: icmp_seq=8 ttl=63 time=161 ms  
64 bytes from 192.168.0.117: icmp_seq=9 ttl=63 time=76.1 ms  
64 bytes from 192.168.0.117: icmp_seq=10 ttl=63 time=102 ms  
64 bytes from 192.168.0.117: icmp_seq=11 ttl=63 time=30.3 ms  
64 bytes from 192.168.0.117: icmp_seq=12 ttl=63 time=129 ms  
64 bytes from 192.168.0.117: icmp_seq=13 ttl=63 time=76.0 ms  
64 bytes from 192.168.0.117: icmp_seq=14 ttl=63 time=102 ms
```

6) ping example.com

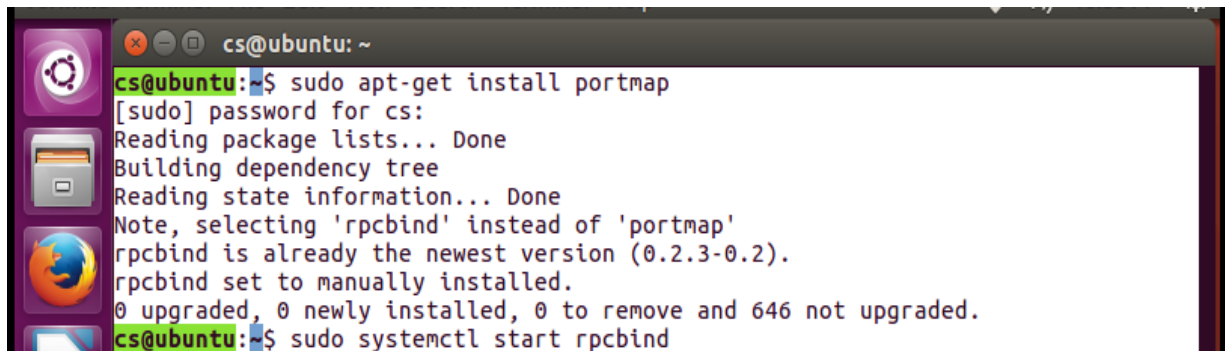


```
pranali@ubuntu: ~  
pranali@ubuntu:~$ ping example.com  
PING example.com (192.168.0.117) 56(84) bytes of data.  
64 bytes from example.com (192.168.0.117): icmp_seq=1 ttl=63 time=79.3 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=2 ttl=63 time=33.4 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=3 ttl=63 time=181 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=4 ttl=63 time=111 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=5 ttl=63 time=42.3 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=6 ttl=63 time=198 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=7 ttl=63 time=226 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=8 ttl=63 time=10.0 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=9 ttl=63 time=368 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=10 ttl=63 time=147 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=11 ttl=63 time=122 ms  
64 bytes from example.com (192.168.0.117): icmp_seq=12 ttl=63 time=172 ms
```

- 7) open browser and then search the file example.com/test.html

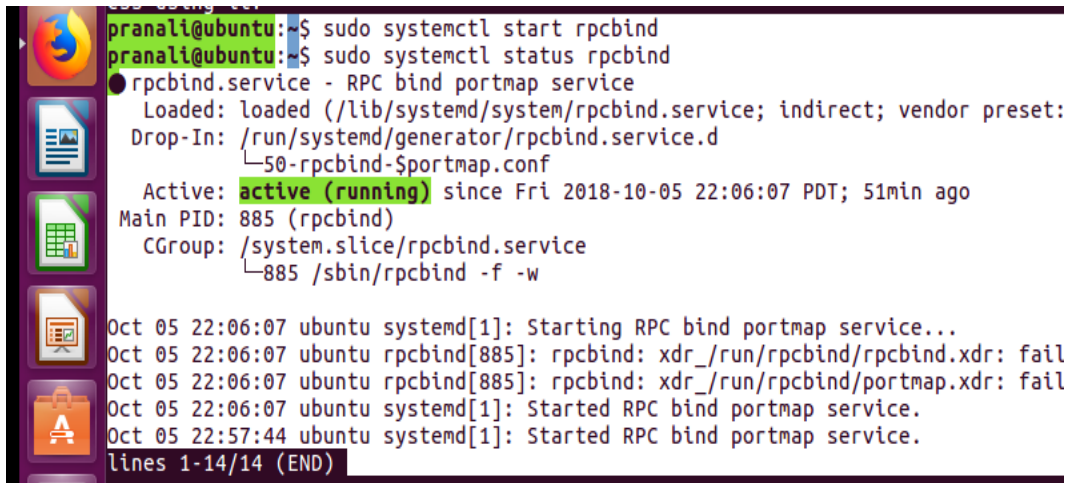


- 8) `sudo apt-get install portmap`



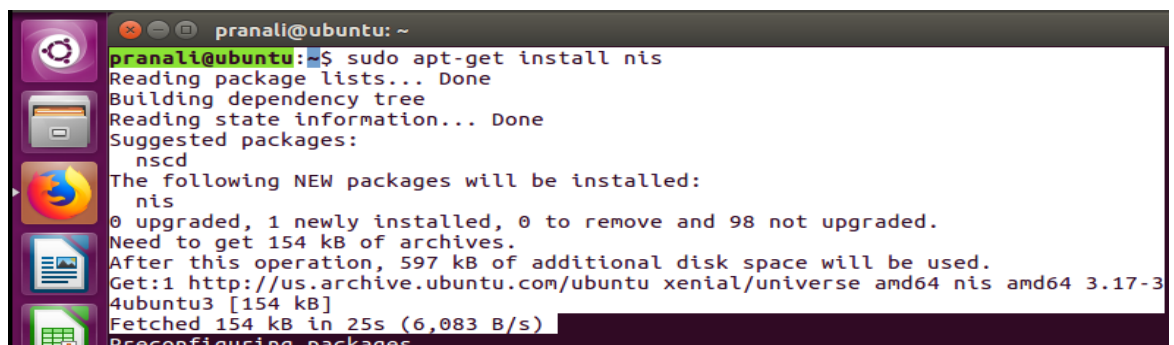
- 9) `sudo systemctl start rpcbind`

- 10) `sudo systemctl status rpcbind`

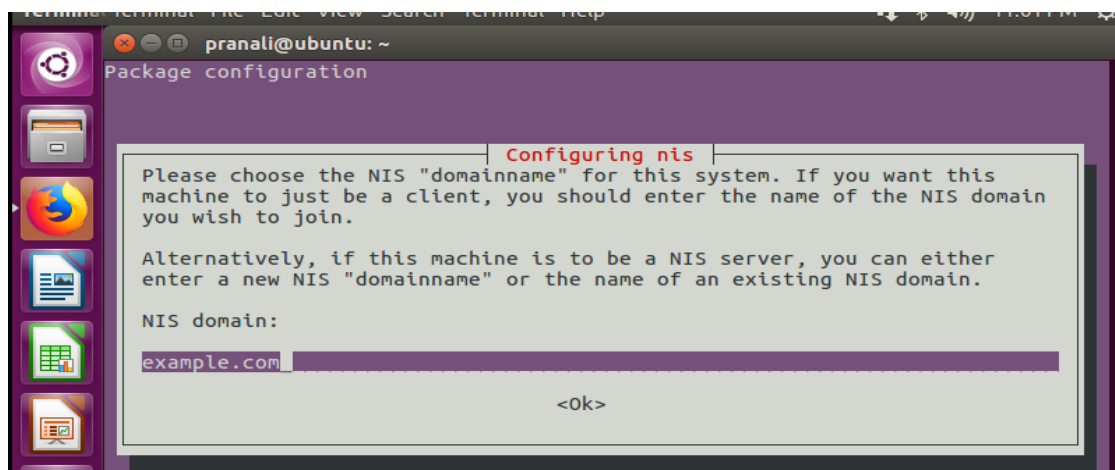


LINUX SERVER ADMINISTRATION

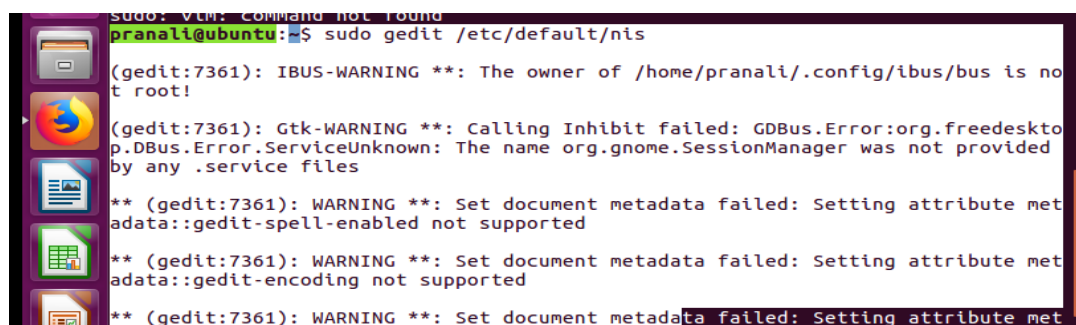
11) sudo apt-get install nis



```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install nis  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
Suggested packages:  
  nscd  
The following NEW packages will be installed:  
  nis  
0 upgraded, 1 newly installed, 0 to remove and 98 not upgraded.  
Need to get 154 kB of archives.  
After this operation, 597 kB of additional disk space will be used.  
Get:1 http://us.archive.ubuntu.com/ubuntu xenial/universe amd64 nis amd64 3.17-3  
4ubuntu3 [154 kB]  
Fetched 154 kB in 25s (6,083 B/s)  
Preconfiguring packages
```



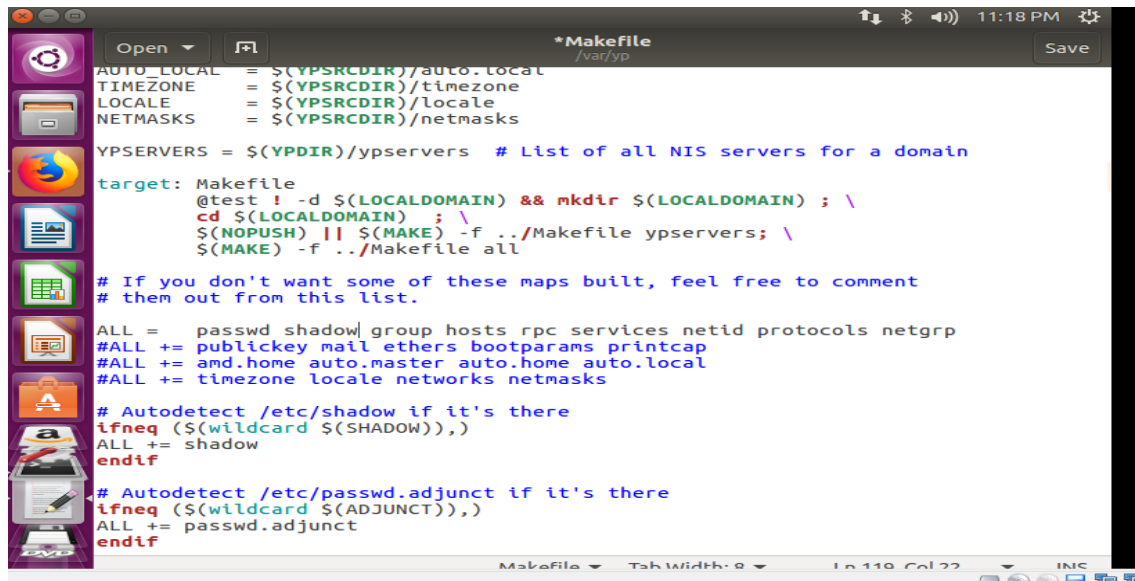
12) sudo gedit /etc/default/nis



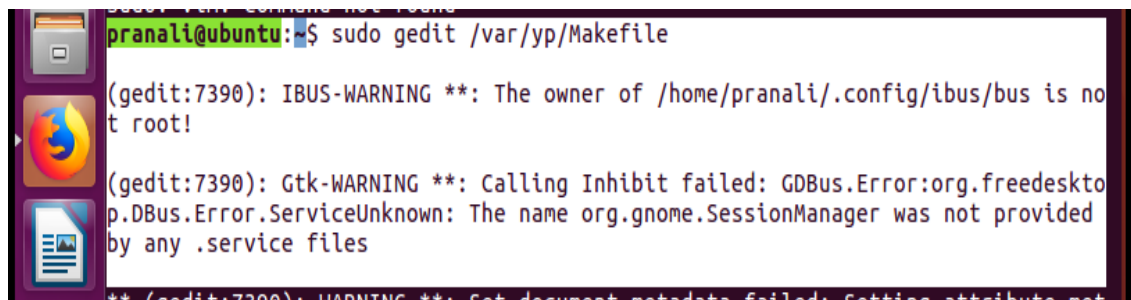
```
sudo: vim: command not found  
pranali@ubuntu:~$ sudo gedit /etc/default/nis  
(gedit:7361): IBUS-WARNING **: The owner of /home/pranali/.config/ibus/bus is not root!  
(gedit:7361): Gtk-WARNING **: Calling Inhibit failed: GDBus.Error:org.freedesktop.DBus.Error.ServiceUnknown: The name org.gnome.SessionManager was not provided by any .service files  
** (gedit:7361): WARNING **: Set document metadata failed: Setting attribute metadata::gedit-spell-enabled not supported  
** (gedit:7361): WARNING **: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported  
** (gedit:7361): WARNING **: Set document metadata failed: Setting attribute met
```

13) sudo gedit /var/yp/Makefile

line number 119 add shadow text

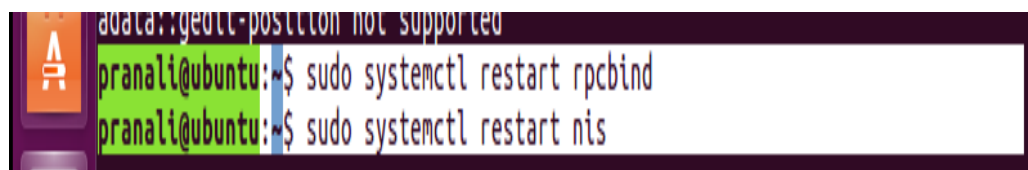


14) sudo gedit /var/yp/Makefile

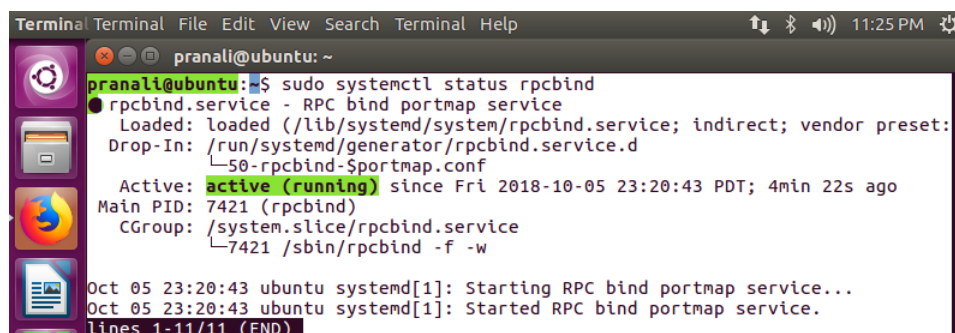


15) sudo systemctl restart rpcbind

16) sudo systemctl restart nis



17) sudo systemctl status rpcbind



18) sudo systemctl status nis

```
pranali@ubuntu: ~$ sudo systemctl status nis
● nis.service - LSB: Start NIS client and server daemons.
   Loaded: loaded (/etc/init.d/nis; bad; vendor preset: enabled)
   Active: active (running) since Fri 2018-10-05 23:23:46 PDT; 1min 50s ago
     Docs: man:systemd-sysv-generator(8)
  Process: 7425 ExecStop=/etc/init.d/nis stop (code=exited, status=0/SUCCESS)
  Process: 7434 ExecStart=/etc/init.d/nis start (code=exited, status=0/SUCCESS)
    CGroup: /system.slice/nis.service
            └─7443 /usr/sbin/ypserv
               7446 /usr/sbin/rpc.yppasswdd -D /etc -e chsh
               7449 /usr/sbin/rpc.ypxfrd
               7457 /usr/sbin/ypbind -broadcast -no-dbus

Oct 05 23:22:43 ubuntu nis[7434]: * ....
Oct 05 23:22:59 ubuntu nis[7434]: * ....
Oct 05 23:23:15 ubuntu nis[7434]: * ....
Oct 05 23:23:31 ubuntu nis[7434]: * ....
Oct 05 23:23:32 ubuntu ypbind[7457]: broadcast: RPC: Timed out.
Oct 05 23:23:46 ubuntu nis[7434]: ...fail!
Oct 05 23:23:46 ubuntu nis[7434]: ...done.
Oct 05 23:23:46 ubuntu systemd[1]: Started LSB: Start NIS client and server daem
Oct 05 23:24:26 ubuntu ypbind[7457]: broadcast: RPC: Timed out.
Oct 05 23:25:20 ubuntu ypbind[7457]: broadcast: RPC: Timed out.

lines 1-22/22 (END)
```

19) sudo /usr/lib/yp/ypinit -m

```
pranali@ubuntu: ~$ sudo /usr/lib/yp/ypinit -m

At this point, we have to construct a list of the hosts which will run NIS
servers. ubuntu is in the list of NIS server hosts. Please continue to add
the names for the other hosts, one per line. When you are done with the
list, type a <control D>.
  next host to add:  ubuntu
  next host to add:
  next host to add:

The current list of NIS servers looks like this:

ubuntu

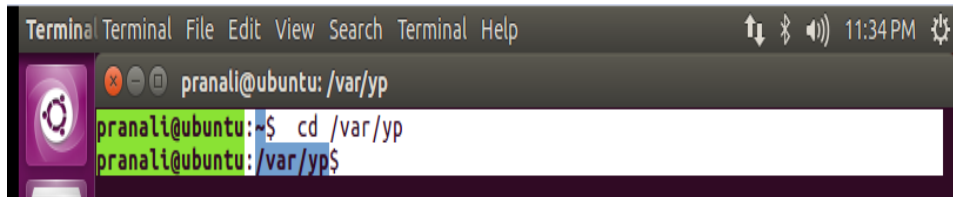
Is this correct? [y/n: y] y
```

20) sudo useradd -d /home/nisusr

21) sudo passwd nisusr

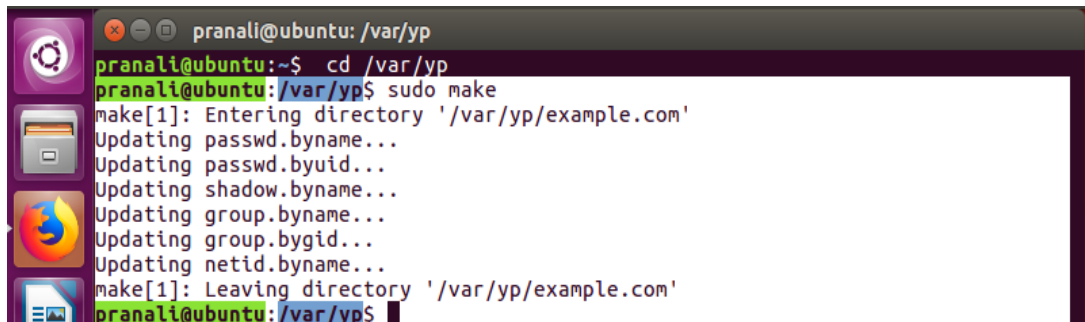
```
pranali@ubuntu:~$ sudo useradd -d /home/nisusr -m nisusr
pranali@ubuntu:~$ sudo passwd nisusr
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
pranali@ubuntu:~$
```

22) cd /var/yp



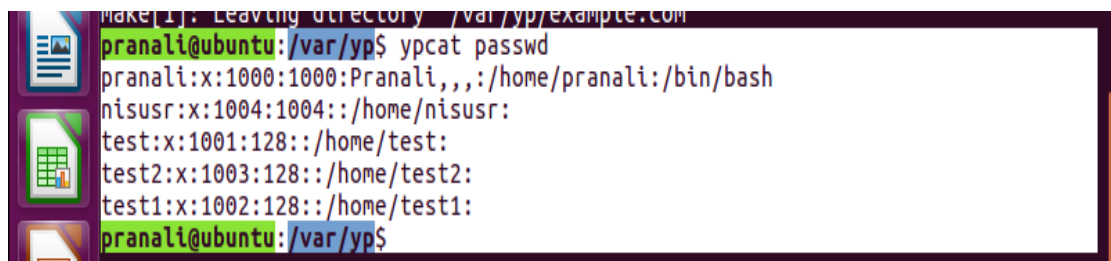
```
pranali@ubuntu: /var/yp
pranali@ubuntu:~$ cd /var/yp
pranali@ubuntu: /var/yp$
```

23) sudo make



```
pranali@ubuntu: /var/yp
pranali@ubuntu:~$ cd /var/yp
pranali@ubuntu: /var/yp$ sudo make
make[1]: Entering directory '/var/yp/example.com'
Updating passwd.byname...
Updating passwd.byuid...
Updating shadow.byname...
Updating group.byname...
Updating group.bygid...
Updating netid.byname...
make[1]: Leaving directory '/var/yp/example.com'
pranali@ubuntu: /var/yp$
```

24) ypcat passwd



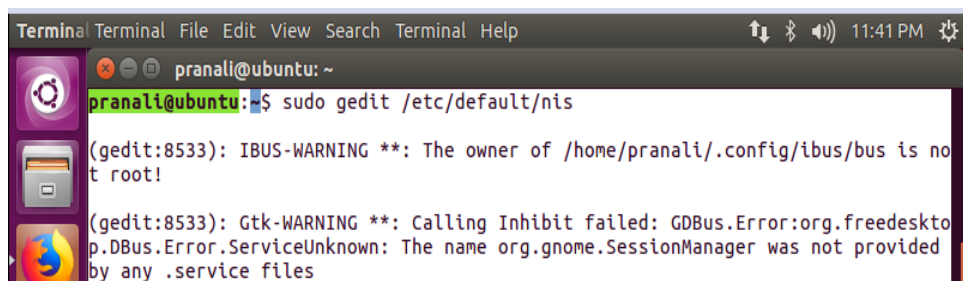
```
pranali@ubuntu: /var/yp$ ypcat passwd
pranali:x:1000:1000:Pranali,,,:/home/pranali:/bin/bash
nisusr:x:1004:1004::/home/nisusr:
test:x:1001:128::/home/test:
test2:x:1003:128::/home/test2:
test1:x:1002:128::/home/test1:
pranali@ubuntu: /var/yp$
```

25) cd ~



```
test1:x:1002:128::/home/test1:
pranali@ubuntu: /var/yp$ cd ~
```

26) sudo gedit /etc/default/nis



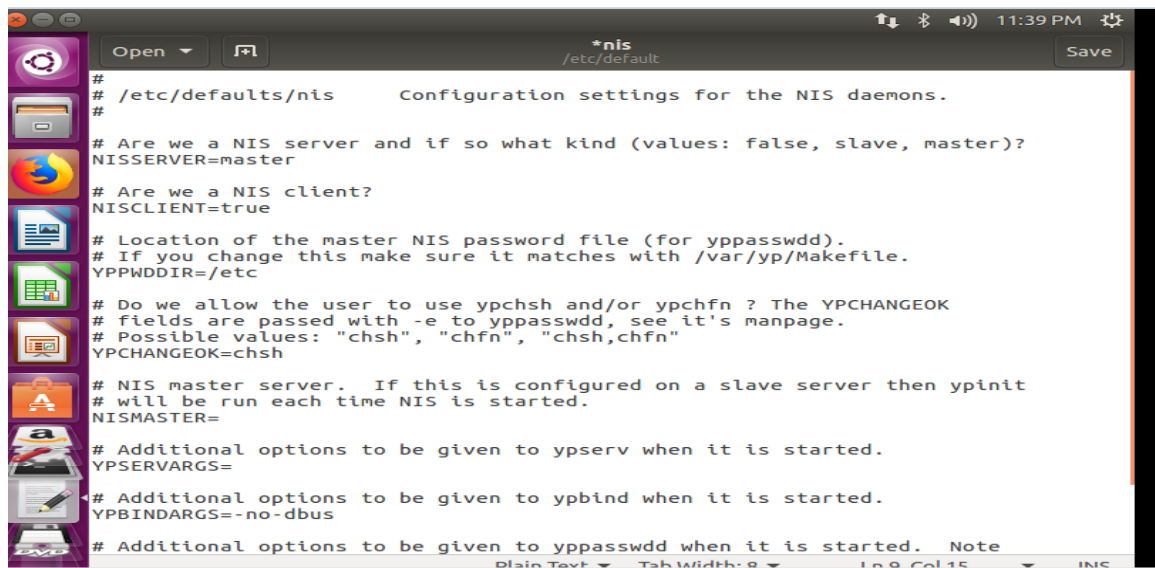
```
pranali@ubuntu: ~
pranali@ubuntu:~$ sudo gedit /etc/default/nis
(gedit:8533): IBUS-WARNING **: The owner of /home/pranali/.config/ibus/bus is not root!
(gedit:8533): Gtk-WARNING **: Calling Inhibit failed: GDBus.Error:org.freedesktop.DBus.Error.ServiceUnknown: The name org.gnome.SessionManager was not provided by any .service files
```

Change the NIS server and NIS client as follow:-

NISSERVER=master

NISCLIENT=true

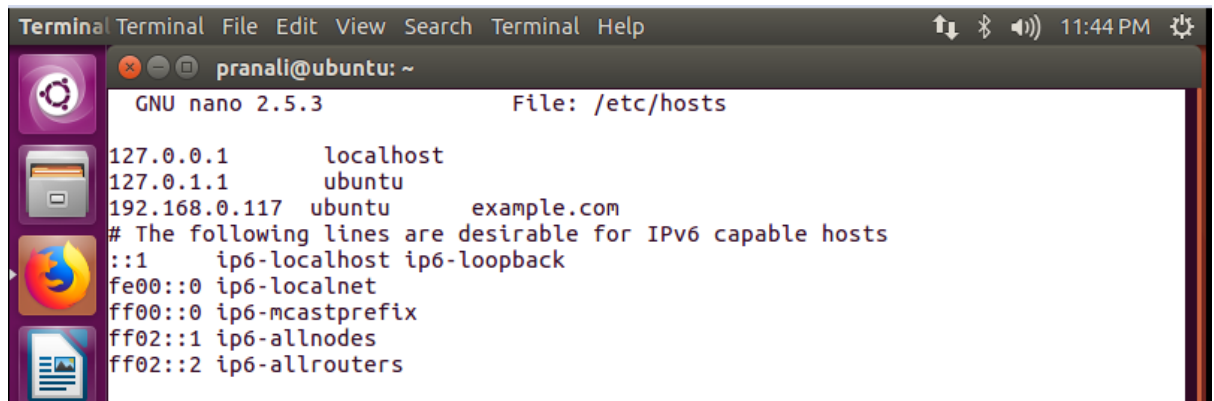
LINUX SERVER ADMINISTRATION



```
#
# /etc/default/nis      Configuration settings for the NIS daemons.
#
# Are we a NIS server and if so what kind (values: false, slave, master)?
NISERVER=master
# Are we a NIS client?
NISCLIENT=true
# Location of the master NIS password file (for yppasswdd).
# If you change this make sure it matches with /var/yp/Makefile.
YPPWDDIR=/etc
# Do we allow the user to use ypchsh and/or ypchfn ? The YPCHANGEOK
# fields are passed with -e to yppasswdd, see it's manpage.
# Possible values: "chsh", "chfn", "chsh,chfn"
YPCHANGEOK=chsh
# NIS master server.  If this is configured on a slave server then ypinit
# will be run each time NIS is started.
NISMASTER=
# Additional options to be given to ypserv when it is started.
YPSERVARGS=
# Additional options to be given to ypbind when it is started.
YPBINDARGS=-no-dbus
# Additional options to be given to yppasswdd when it is started.  Note
```

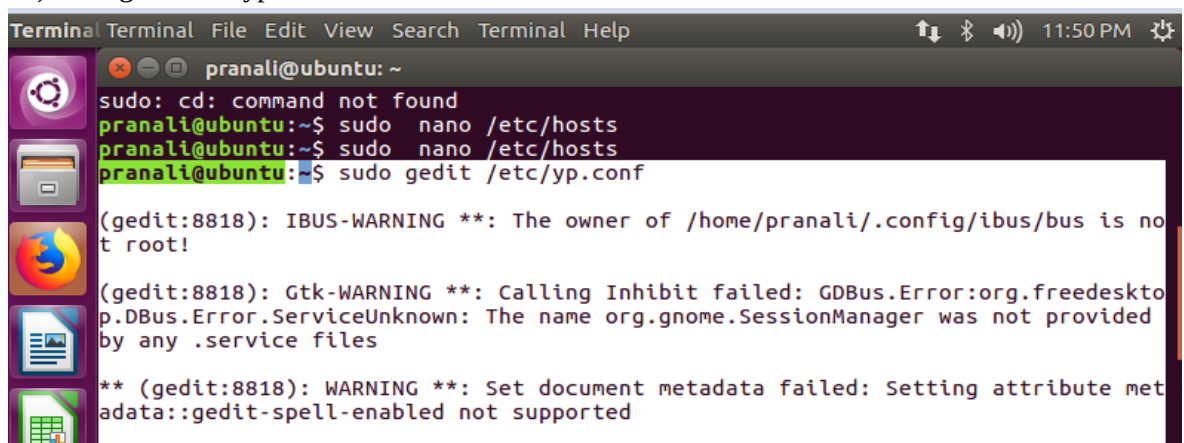
Add following line in line number 5

Ip address of system ubuntu example.com



```
pranali@ubuntu: ~
GNU nano 2.5.3      File: /etc/hosts
127.0.0.1      localhost
127.0.1.1      ubuntu
192.168.0.117  ubuntu      example.com
# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

27) `sudo gedit /etc/yp.conf`



```
pranali@ubuntu: ~
sudo: cd: command not found
pranali@ubuntu:~$ sudo nano /etc/hosts
pranali@ubuntu:~$ sudo nano /etc/hosts
pranali@ubuntu:~$ sudo gedit /etc/yp.conf

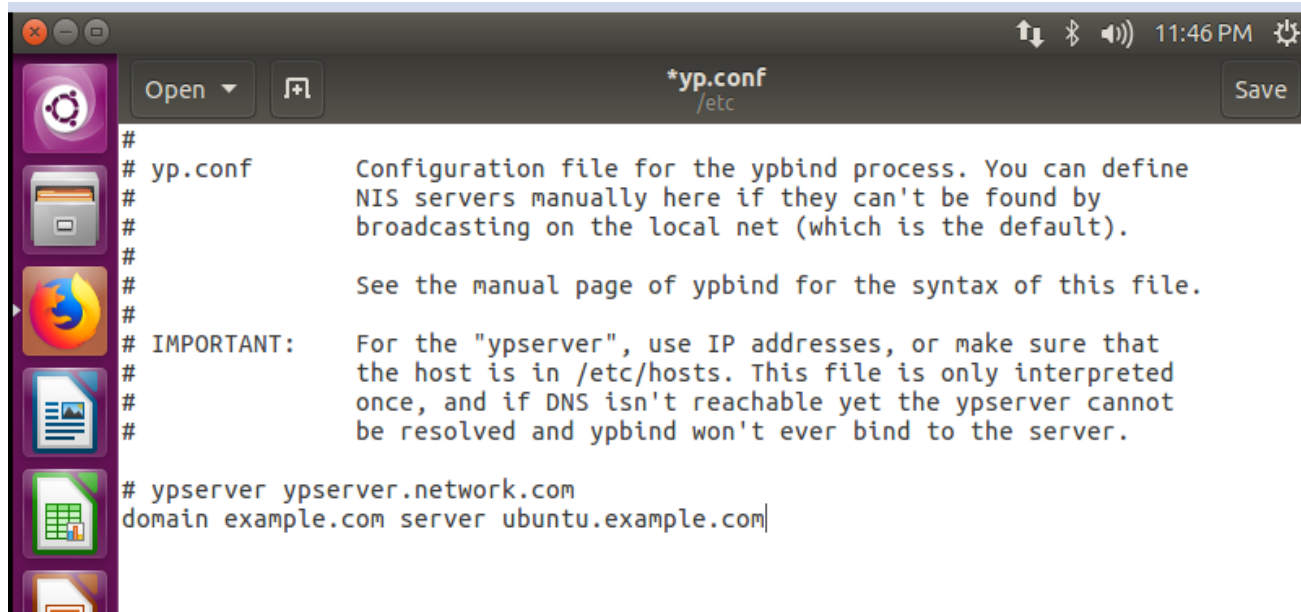
(gedit:8818): IBUS-WARNING **: The owner of /home/pranali/.config/ibus/bus is no
t root!

(gedit:8818): Gtk-WARNING **: Calling Inhibit failed: GDBus.Error:org.freedesktop
.DBus.Error.ServiceUnknown: The name org.gnome.SessionManager was not provided
by any .service files

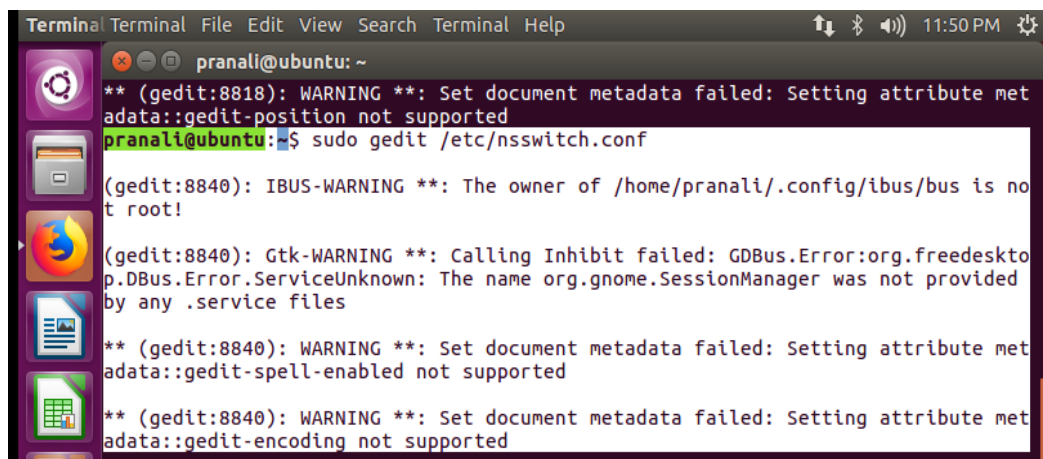
** (gedit:8818): WARNING **: Set document metadata failed: Setting attribute met
adata::gedit-spell-enabled not supported
```

Add following lines

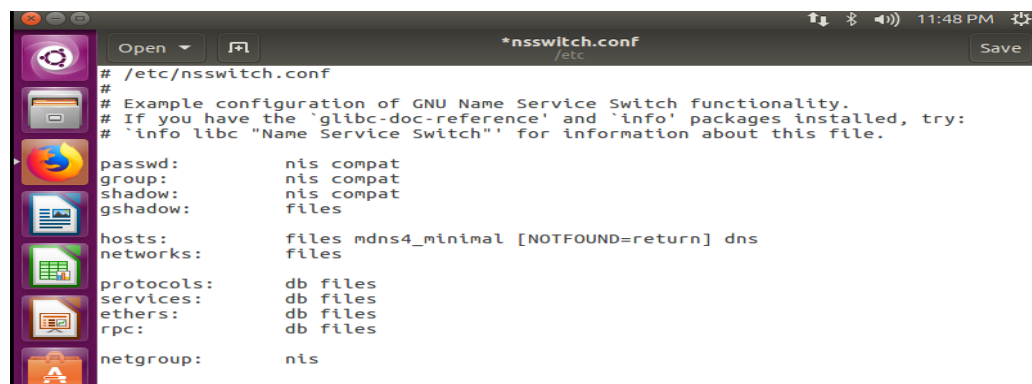
domain example.com server ubuntu.example.com



28) `sudo gedit /etc/nsswitch.conf`



Add nis in every compat word



29) sudo chmod 777 /home



```
Try 'chmod --help' for more information.  
pranali@ubuntu:~$ sudo chmod 777 /home  
pranali@ubuntu:~$ ypcat passwd  
pranali:x:1000:1000:Pranali,,,:/home/pranali:/bin/bash  
nisusr:x:1004:1004::/home/nisusr:  
test:x:1001:128::/home/test:  
test2:x:1003:128::/home/test2:  
test1:x:1002:128::/home/test1:  
pranali@ubuntu:~$
```

30) reboot



```
pranali@ubuntu: ~  
pranali@ubuntu:~$ reboot
```

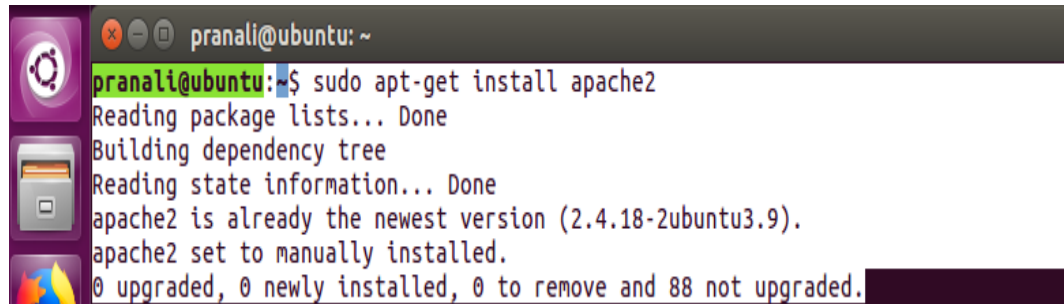
PRACTICAL NO:-09

AIM:-

Install MySQL to configure database server, Install phpMyAdmin to operate MySQL on web browser from Clients.

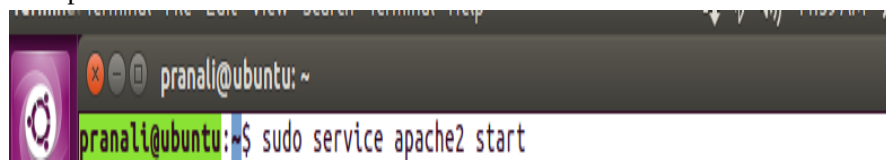
COMMAND:-

- 1) `sudo apt-get install apache2`



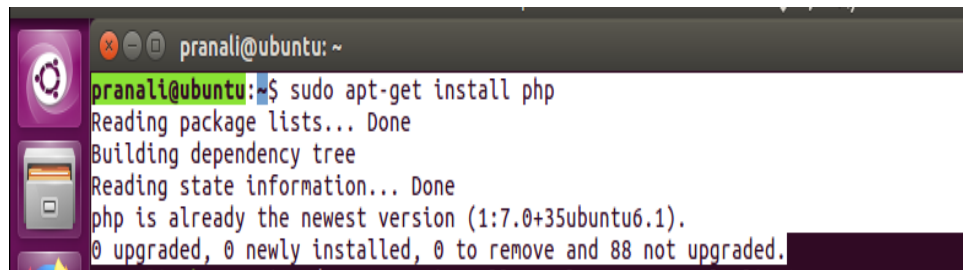
```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install apache2  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
apache2 is already the newest version (2.4.18-2ubuntu3.9).  
apache2 set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
```

- 2) `sudo service apache2 start`



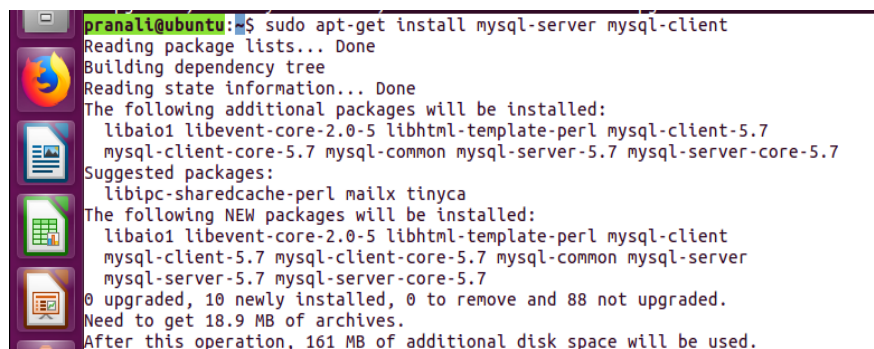
```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo service apache2 start
```

- 3) `sudo apt-get install php`



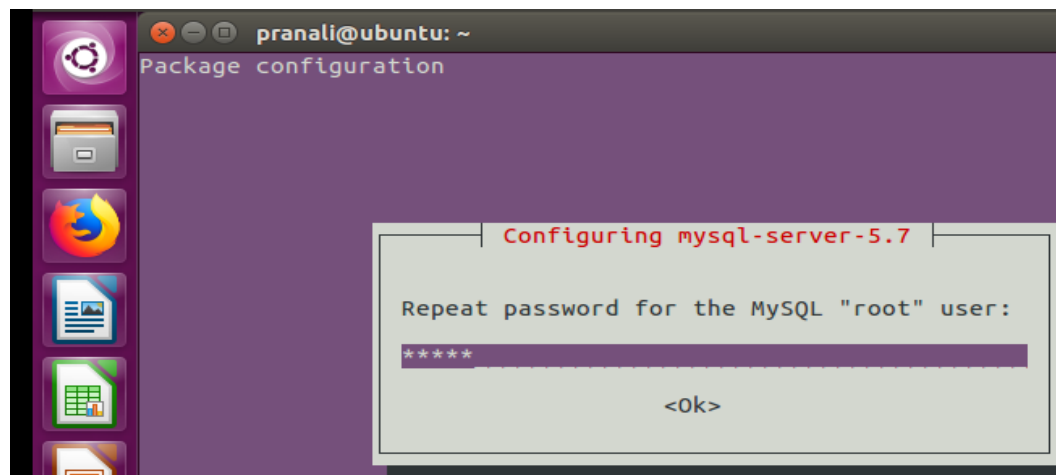
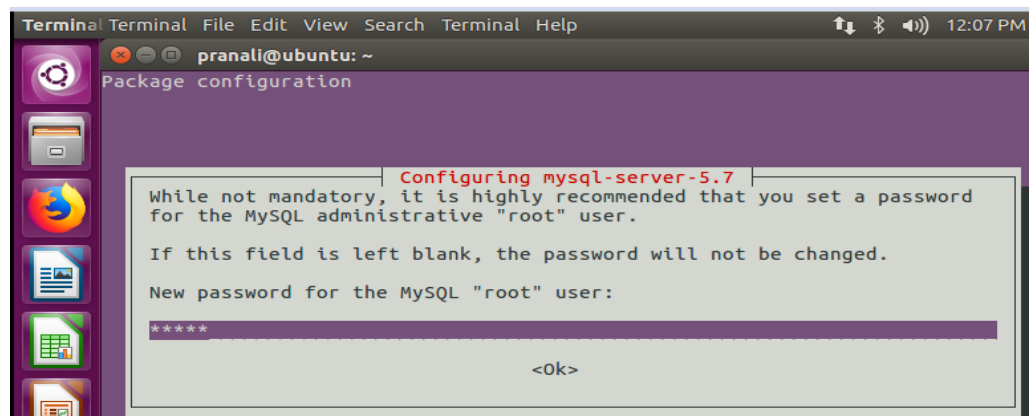
```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install php  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
php is already the newest version (1:7.0+35ubuntu6.1).  
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
```

- 4) `sudo apt-get install mysql-server mysql-client`

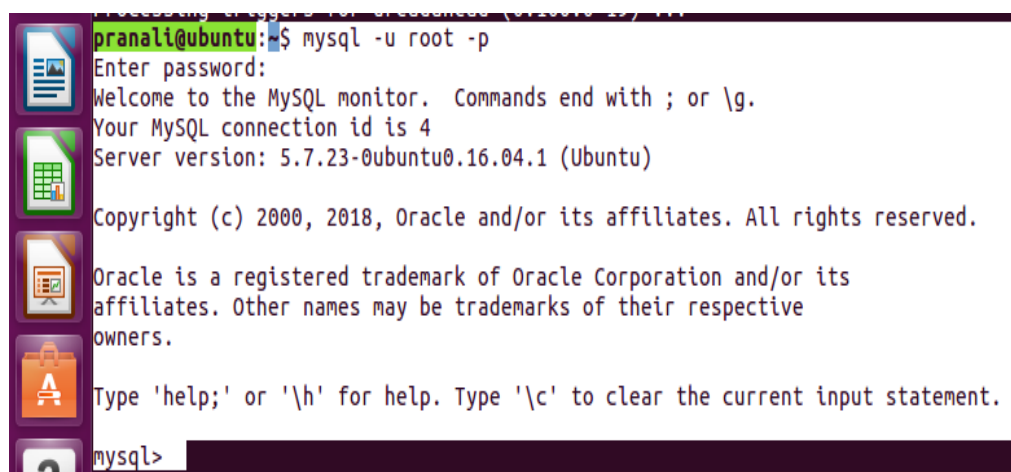


```
pranali@ubuntu:~$ sudo apt-get install mysql-server mysql-client  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  libaio1 libevent-core-2.0-5 libhtml-template-perl mysql-client-5.7  
  mysql-client-core-5.7 mysql-common mysql-server-5.7 mysql-server-core-5.7  
Suggested packages:  
  libipc-sharedcache-perl mailx tinyca  
The following NEW packages will be installed:  
  libaio1 libevent-core-2.0-5 libhtml-template-perl mysql-client  
  mysql-client-5.7 mysql-client-core-5.7 mysql-common mysql-server  
  mysql-server-5.7 mysql-server-core-5.7  
0 upgraded, 10 newly installed, 0 to remove and 88 not upgraded.  
Need to get 18.9 MB of archives.  
After this operation, 161 MB of additional disk space will be used.
```

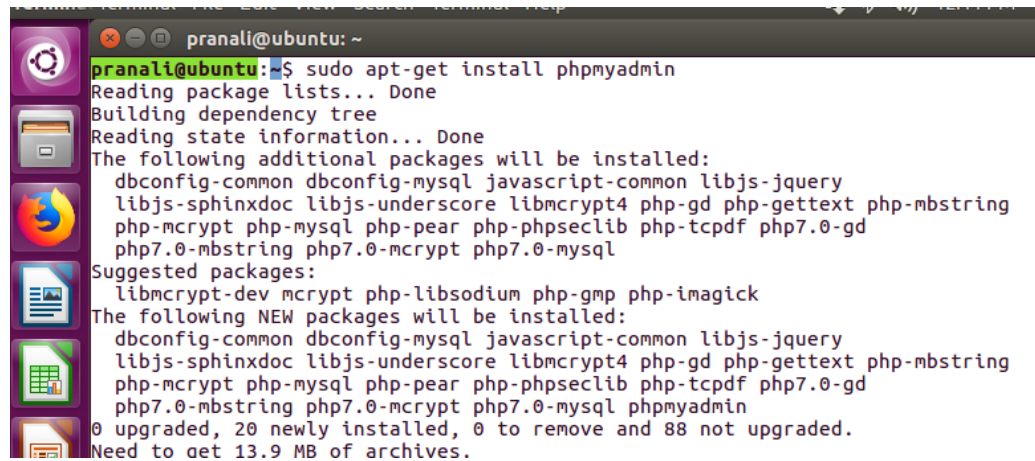
- 5) Default user is root then set the password 12345.



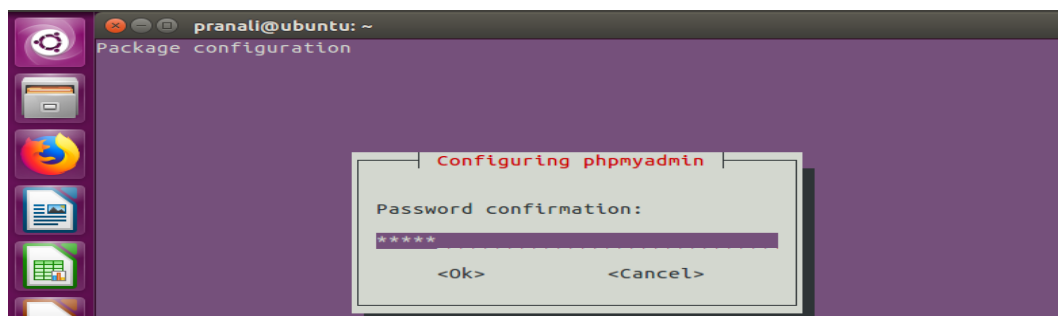
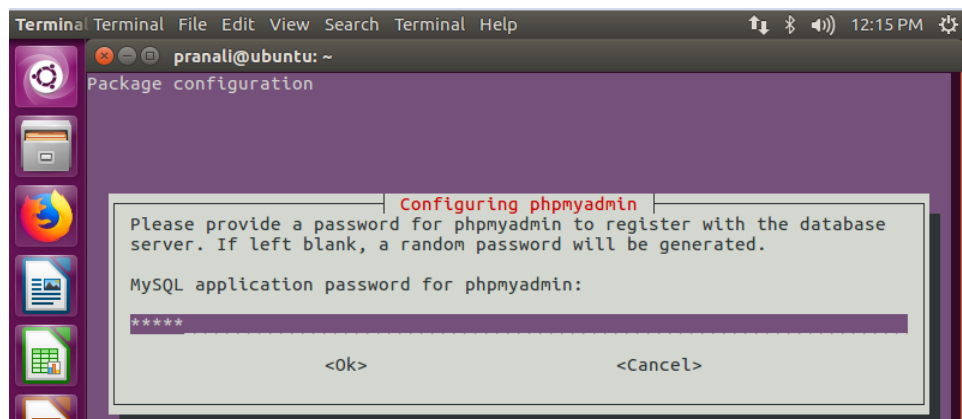
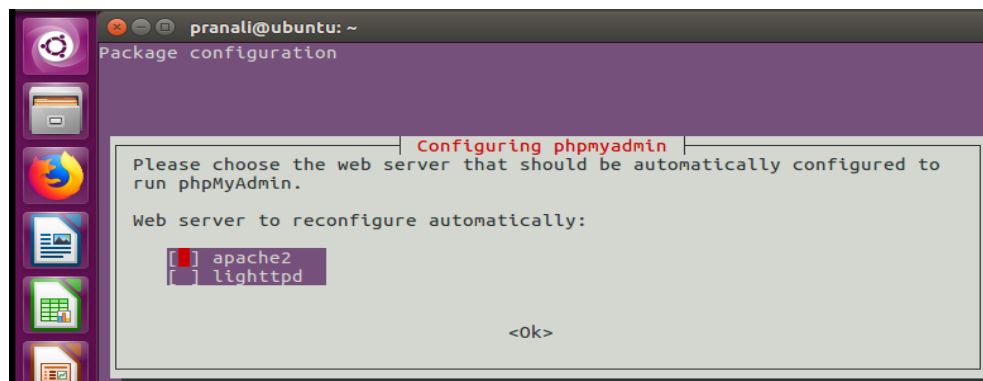
6) Mysql -u root -p



7) Sudo apt-get install phpmyadmin



```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install phpmyadmin  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  dbconfig-common dbconfig-mysql javascript-common libjs-jquery  
  libjs-sphinxdoc libjs-underscore libmcrypt4 php-gd php-gettext php-mbstring  
  php-mcrypt php-mysql php-pear php-phpseclib php-tcpdf php7.0-gd  
  php7.0-mbstring php7.0-mcrypt php7.0-mysql  
Suggested packages:  
  libmcrypt-dev mcrypt php-libsodium php-gmp php-imagick  
The following NEW packages will be installed:  
  dbconfig-common dbconfig-mysql javascript-common libjs-jquery  
  libjs-sphinxdoc libjs-underscore libmcrypt4 php-gd php-gettext php-mbstring  
  php-mcrypt php-mysql php-pear php-phpseclib php-tcpdf php7.0-gd  
  php7.0-mbstring php7.0-mcrypt php7.0-mysql phpmyadmin  
0 upgraded, 20 newly installed, 0 to remove and 88 not upgraded.  
Need to get 13.9 MB of archives.
```



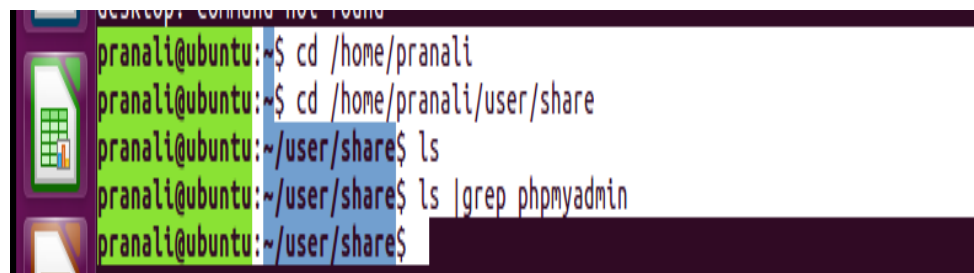
LINUX SERVER ADMINISTRATION

- 8) `ls`
- 9) `cd /var/www/html`



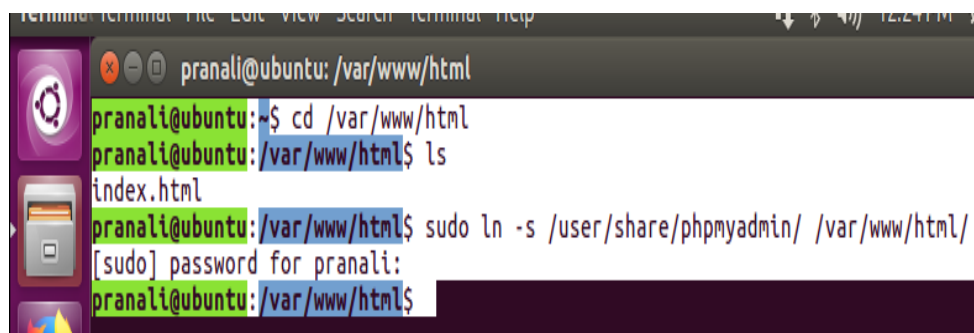
```
pranali@ubuntu:~$ ls
Desktop  Downloads  Music  Public  Videos
Documents  examples.desktop  Pictures  Templates
pranali@ubuntu:~$ locate phpmyadmin
pranali@ubuntu:~$ cd /var/www/html
pranali@ubuntu:/var/www/html$ ls
index.html
```

- 10) `/user/share $ls`
- 11) `/user/share $ls | grep phpmyadmin`



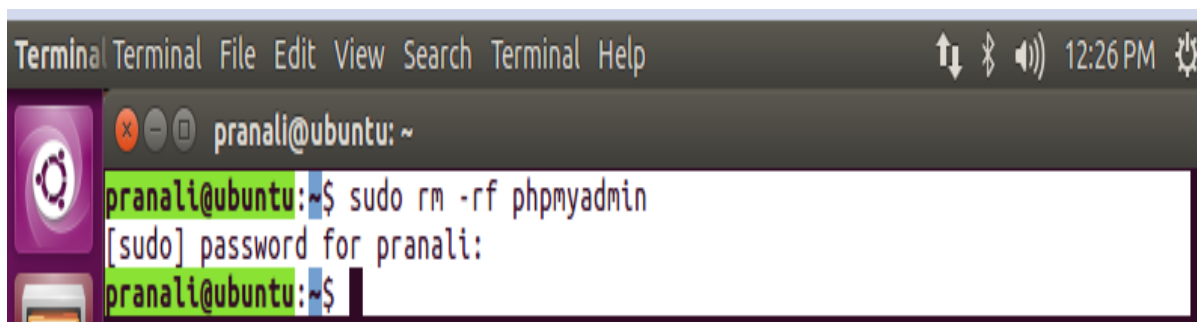
```
pranali@ubuntu:~$ cd /home/pranali
pranali@ubuntu:~/user/share$ ls
pranali@ubuntu:~/user/share$ ls | grep phpmyadmin
```

- 12) `Cd /var/www/html`
- 13) `Cd /var/www/html $ sudo ln -s /user/share/phpmyadmin/ /var/www/html/`



```
pranali@ubuntu:/var/www/html$ cd /var/www/html
pranali@ubuntu:/var/www/html$ ls
index.html
pranali@ubuntu:/var/www/html$ sudo ln -s /user/share/phpmyadmin/ /var/www/html/
[sudo] password for pranali:
pranali@ubuntu:/var/www/html$
```

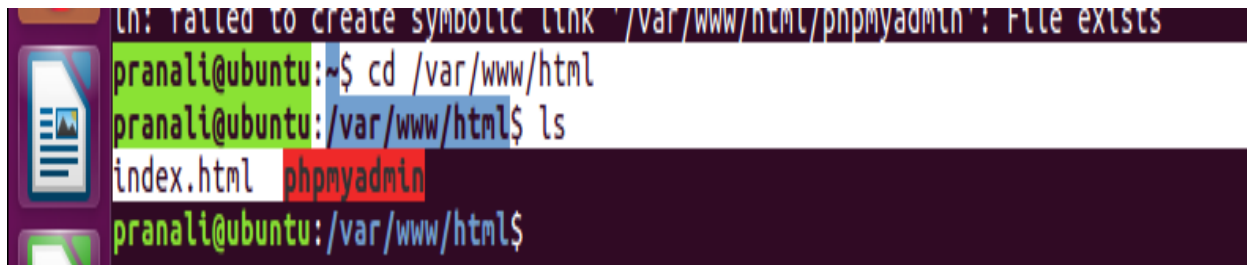
- 14) Refresh the browser localhost/phpmyadmin
- 15) If forbidden error is occur then
`Sudo rm -rf phpmyadmin`



```
pranali@ubuntu:~$ sudo rm -rf phpmyadmin
[sudo] password for pranali:
pranali@ubuntu:~$
```

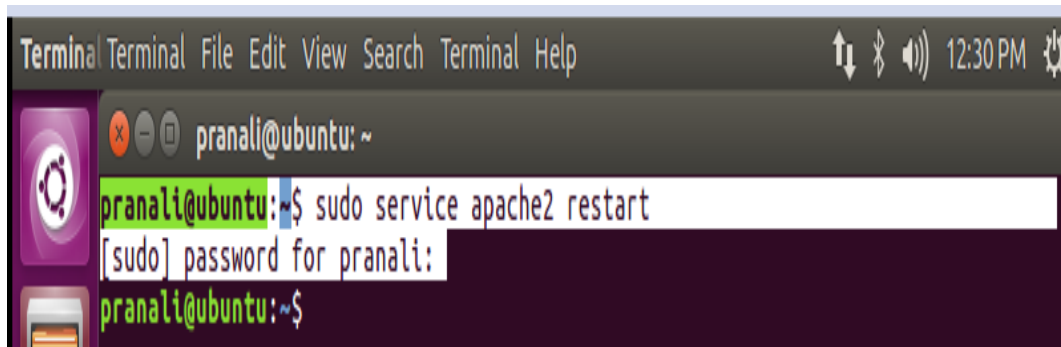
16) `Cd /var/www/html`

17) `Cd /var/www/html $ ls`

A terminal window with a dark purple background. The prompt is `pranali@ubuntu:~$`. The user enters `cd /var/www/html`. The prompt changes to `pranali@ubuntu:/var/www/html$`. The user enters `ls`. The output shows `index.html` and `phpmyadmin`. The prompt returns to `pranali@ubuntu:/var/www/html$`.

```
ln: failed to create symbolic link '/var/www/html/phpmyadmin': File exists
pranali@ubuntu:~$ cd /var/www/html
pranali@ubuntu:/var/www/html$ ls
index.html  phpmyadmin
pranali@ubuntu:/var/www/html$
```

18) `Sudo service apache2 restart`

A terminal window titled "Terminal" with a menu bar (Terminal, File, Edit, View, Search, Terminal, Help) and system icons (up/down arrows, Bluetooth, volume, 12:30 PM, settings). The prompt is `pranali@ubuntu: ~`. The user enters `sudo service apache2 restart`. The prompt changes to `[sudo] password for pranali:`. The user enters their password. The prompt returns to `pranali@ubuntu:~$`.

```
Terminal Terminal File Edit View Search Terminal Help 12:30 PM
pranali@ubuntu: ~
pranali@ubuntu:~$ sudo service apache2 restart
[sudo] password for pranali:
pranali@ubuntu:~$
```

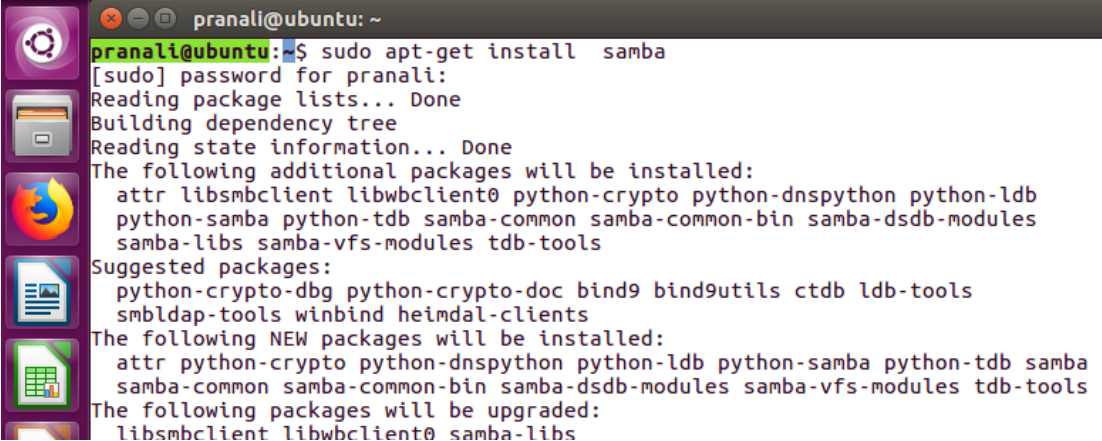
PRACTICAL NO:-10

AIM:-

Install Samba to share folders or files between Windows and Linux

COMMAND:-

- 1) Sudo apt-get install samba



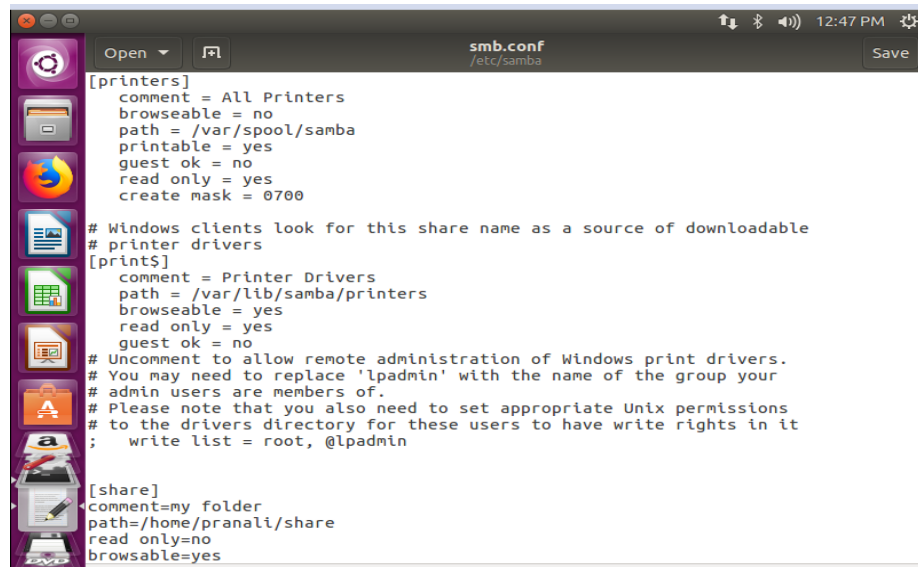
```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo apt-get install samba  
[sudo] password for pranali:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  attr libsmclient libwbclient0 python-crypto python-dnspython python-ldb  
  python-samba python-tdb samba-common samba-common-bin samba-dsdb-modules  
  samba-libs samba-vfs-modules tdb-tools  
Suggested packages:  
  python-crypto-dbg python-crypto-doc bind9 bind9utils ctdb ldb-tools  
  smbldap-tools winbind heimdal-clients  
The following NEW packages will be installed:  
  attr python-crypto python-dnspython python-ldb python-samba python-tdb samba  
  samba-common samba-common-bin samba-dsdb-modules samba-vfs-modules tdb-tools  
The following packages will be upgraded:  
  libsmclient libwbclient0 samba-libs
```

- 2) Create folder in home directory
- 3) Sudo gedit /etc/samba/smb.conf

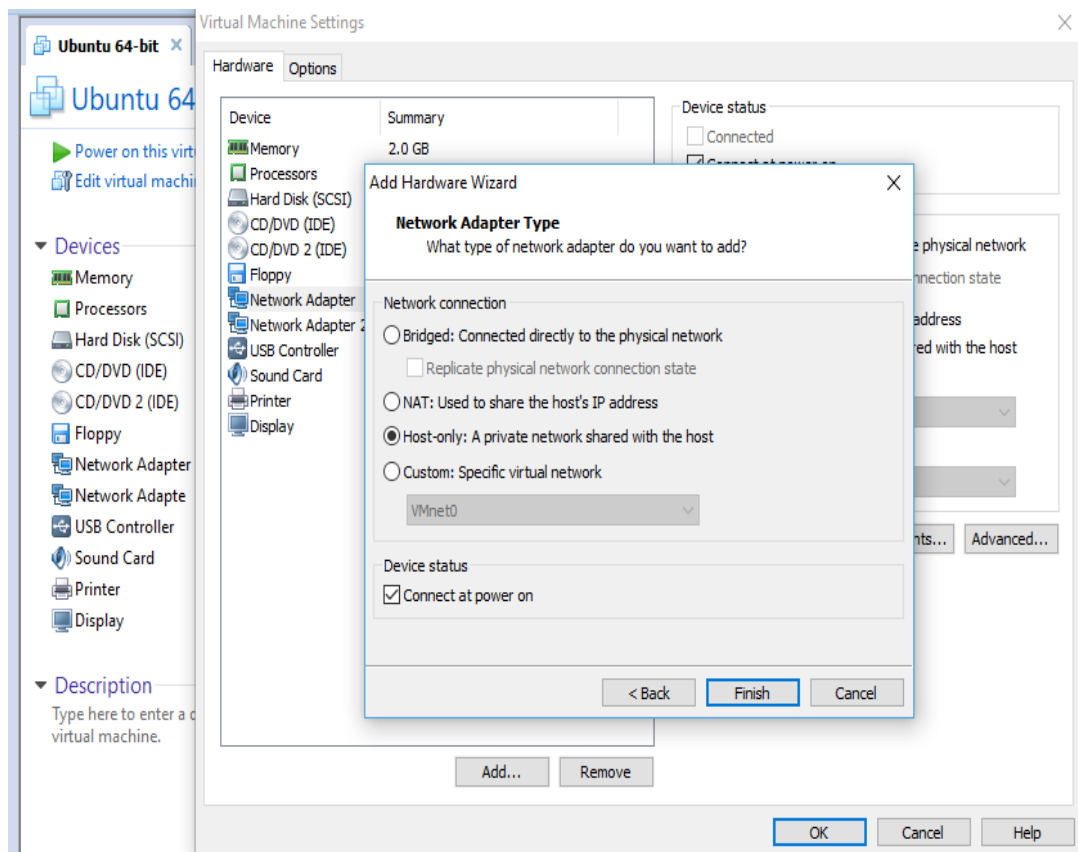


```
pranali@ubuntu: ~  
pranali@ubuntu:~$ sudo gedit /etc/samba/smb.conf  
[sudo] password for pranali:
```

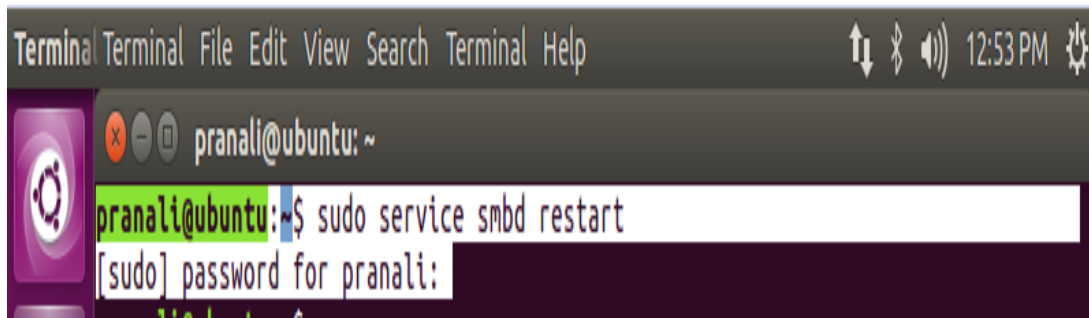
[folder_name]
Comment=my folder
Read only=no
Browsable=yes
Path=/home/Pranali/share



- 4) Then shut down the system and change the network adapter setting NAT into HOST only.

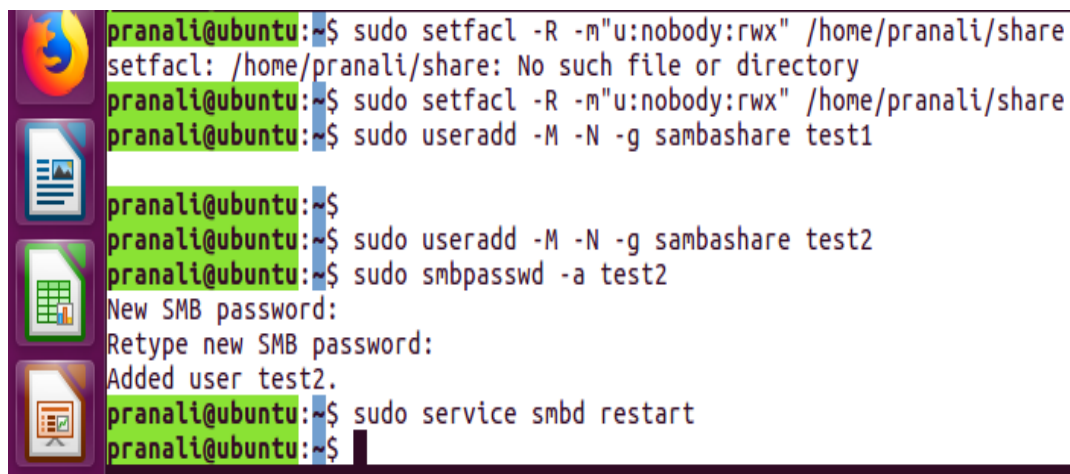


- 5) Sudo service smb restart



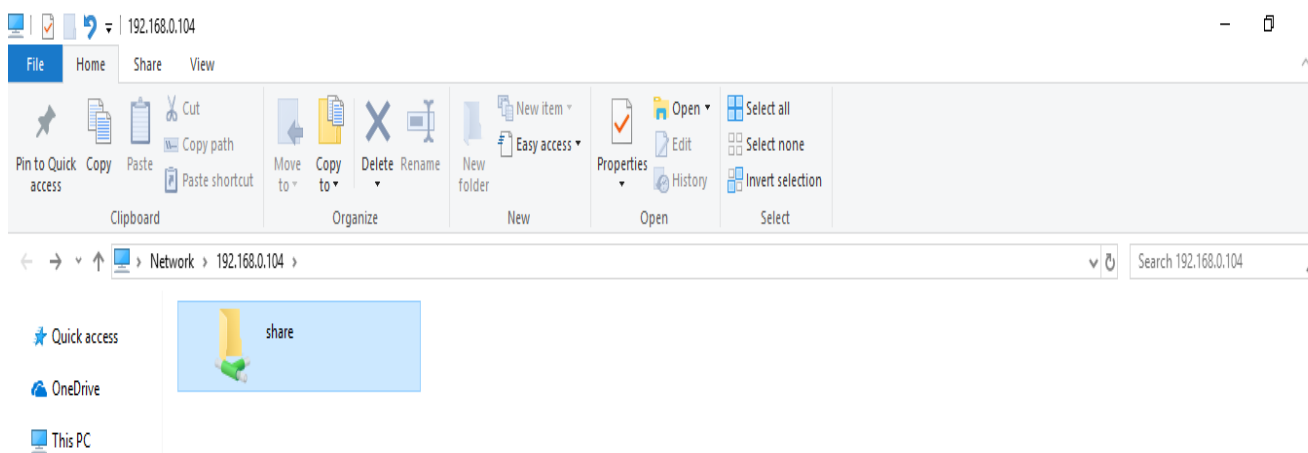
```
Terminal Terminal File Edit View Search Terminal Help
pranali@ubuntu: ~
pranali@ubuntu:~$ sudo service smb restart
[sudo] password for pranali:
```

- 6) Then check the folder is in windows system
7) Window+R :- ip address of ubuntu system.
8) If the folder is lock then
9) Sudo setfacl -R -m"u:nobody:rw" /home/Pranali/user
10) And the user :- sudo useradd -M -N -g sambashare test
11) Set the password of user:- sudo smbpasswd -a test
12) Sudo service smb restart



```
pranali@ubuntu:~$ sudo setfacl -R -m"u:nobody:rw" /home/pranali/share
setfacl: /home/pranali/share: No such file or directory
pranali@ubuntu:~$ sudo setfacl -R -m"u:nobody:rw" /home/pranali/share
pranali@ubuntu:~$ sudo useradd -M -N -g sambashare test1
pranali@ubuntu:~$
pranali@ubuntu:~$ sudo useradd -M -N -g sambashare test2
pranali@ubuntu:~$ sudo smbpasswd -a test2
New SMB password:
Retype new SMB password:
Added user test2.
pranali@ubuntu:~$ sudo service smb restart
pranali@ubuntu:~$
```

- 13) Check the folder



14) Empty

